```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[NG_FFCutsFits,CutFitFileNG] = recover_selected_cuts('NG_FFCutsFits', ...
    'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts');
```

```
fn = fieldnames(NG_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off110NGminFFns'
```

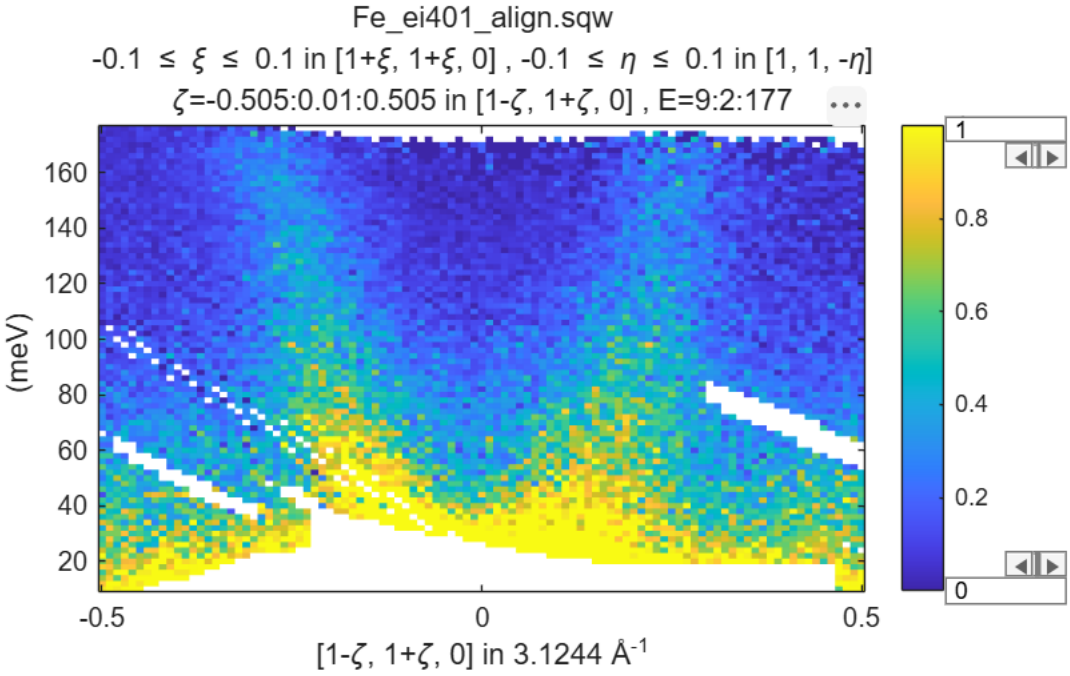
```
selectCuts =
'Ei400_off110NGminFFns'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
        contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei400_off110NGminFFns
```

```
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 1; keep_figure;
end
```




```
dir_name  = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;
```

```
NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
******************************
**** En = 45±5 going to 165
******************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.38964s    Reduced Chi^2 = 11.7841
 Free parameter values:
           10         0.7917             34         0.6634
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 1.5694s    Reduced Chi^2 = 1.9996      Levenberg-Marquardt = 0.1
 Free parameter values:
        15.25         0.7485          20.85         0.4875
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.1625s    Reduced Chi^2 = 1.2417      Levenberg-Marquardt = 0.01
```

```
 Free parameter values:
        1.334         0.5544          22.77         0.4831
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 4.4546s    Reduced Chi^2 = 1.0727      Levenberg-Marquardt = 0.001
 Free parameter values:
        4.085         0.6452           23.7         0.4738
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 5.5387s    Reduced Chi^2 = 1.063      Levenberg-Marquardt = 0.0001
 Free parameter values:
        4.998         0.6597          23.38         0.4713
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 6.6371s    Reduced Chi^2 = 1.0563      Levenberg-Marquardt = 1e-05
 Free parameter values:
        4.768         0.6595          23.27         0.4714
------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 7.761s    Reduced Chi^2 = 1.062      Levenberg-Marquardt = 1e-06
 Free parameter values:
        4.533         0.6529          23.31         0.4723
 Total time = 8.0487s    Reduced Chi^2 = 1.1022      Levenberg-Marquardt = 1e-05
 Free parameter values:
        4.533         0.6529          23.31         0.4723
 Total time = 8.3269s    Reduced Chi^2 = 1.0724      Levenberg-Marquardt = 0.0001
 Free parameter values:
        4.534         0.6529          23.31         0.4723
 Total time = 8.6185s    Reduced Chi^2 = 1.0689      Levenberg-Marquardt = 0.01
 Free parameter values:
        4.542         0.6532          23.31         0.4723
 Total time = 8.8955s    Reduced Chi^2 = 1.0652      Levenberg-Marquardt = 1
 Free parameter values:
        4.686         0.6572           23.3         0.4716
 Total time = 9.1753s    Reduced Chi^2 = 1.0693      Levenberg-Marquardt = 100
 Free parameter values:
         4.76         0.6592          23.27         0.4714
 *** No improvement in chi-squared over previous iteration ***
 Total time = 9.1824s    Reduced Chi^2 = 1.0563
 Free parameter values:
        4.768         0.6595          23.27         0.4714
------------------------------------------------------------------------------
Fit converged:

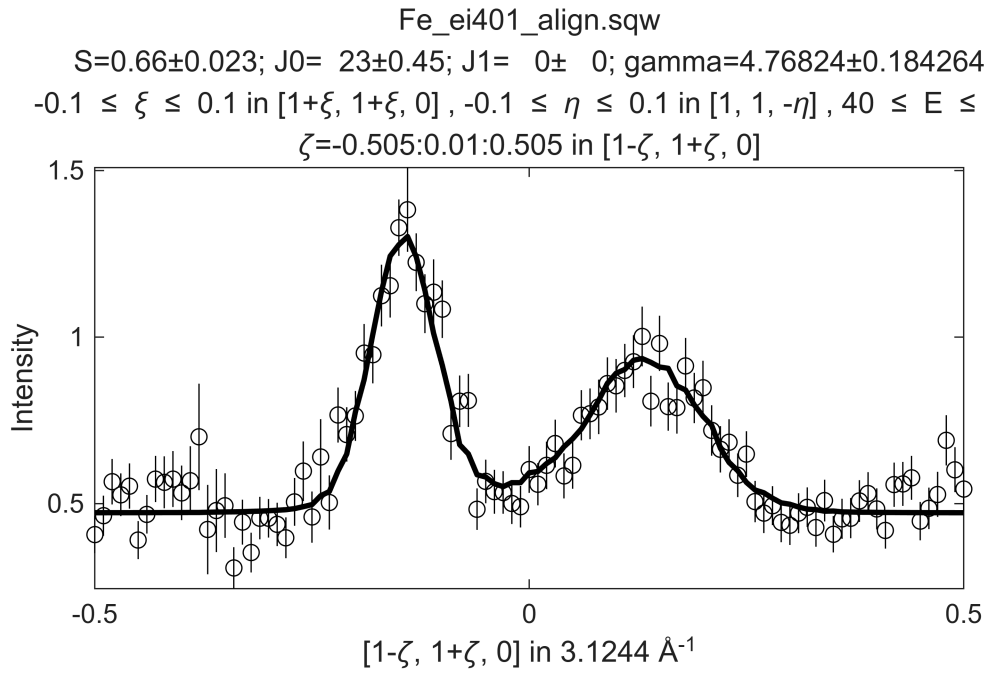
Parameter values (free parameters with error estimates):
---------------------------------------------------------

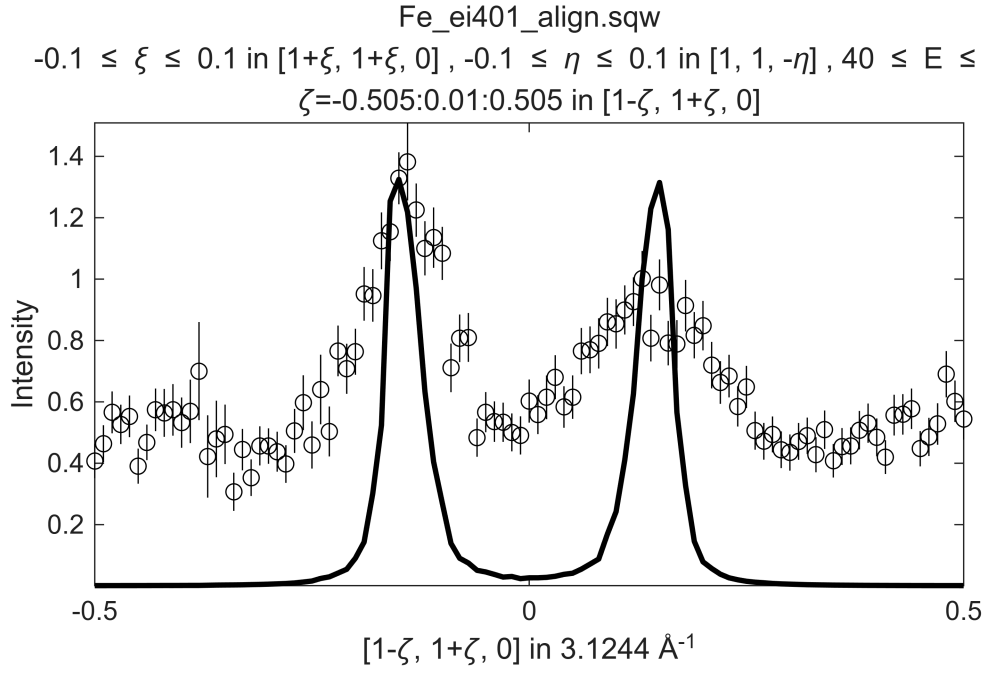
Foreground parameter values:
----------------------------
    1             1
    2             8
    3         4.768   +/-   0.1843
    4        0.6595   +/-   0.02334
    5             0
    6         23.27   +/-   0.4533
    7             0
    8             0
    9             0
   10             0


Background parameter values:
----------------------------
    1        0.4714   +/-   0.008725
```

```
    2              0
    3              0


Covariance matrix for free parameters:
-------------------------------------
    1.0000   -0.7237   -0.6607   -0.0097
   -0.7237    1.0000    0.1387   -0.5466
   -0.6607    0.1387    1.0000    0.0626
   -0.0097   -0.5466    0.0626    1.0000
```






```
*****************************
**** En = 55±5 going to 165
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.32156s    Reduced Chi^2 = 8.6801
 Free parameter values:
         4.768         0.6595          23.27         0.5328
--------------------------------------------------------------------------------
```

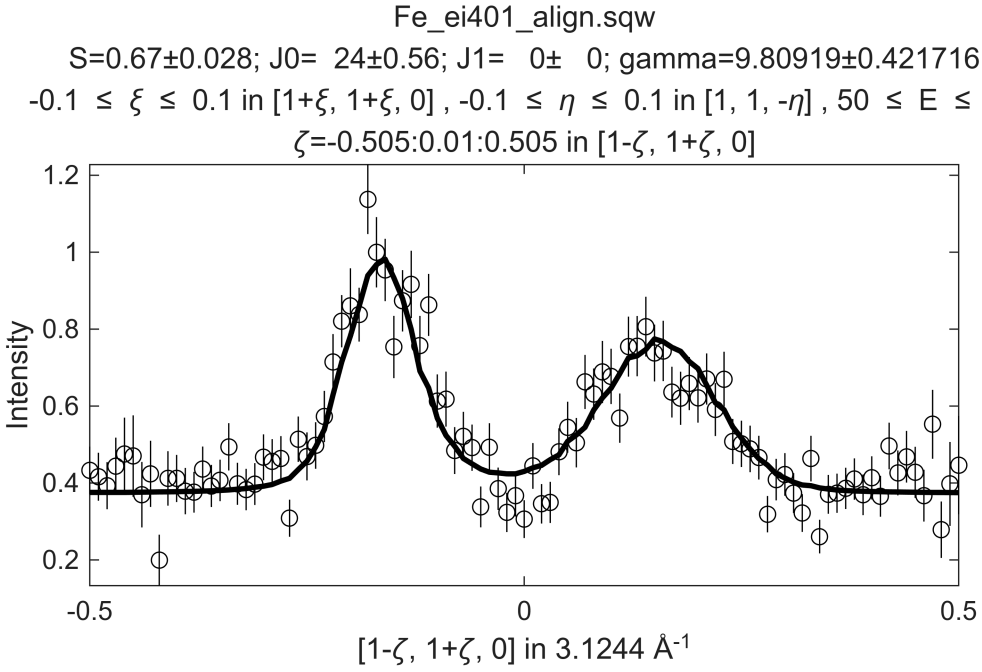
```
Iteration = 1
------------------
 Total time = 1.4529s    Reduced Chi^2 = 1.376      Levenberg-Marquardt = 0.1
 Free parameter values:
         7.837         0.6129          23.57         0.3856
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 2.563s    Reduced Chi^2 = 1.3465      Levenberg-Marquardt = 0.01
 Free parameter values:
         10.13         0.6723          23.85         0.3711
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 3.655s    Reduced Chi^2 = 1.3655      Levenberg-Marquardt = 0.001
 Free parameter values:
         9.792         0.6679           23.9         0.3722
 Total time = 3.9348s    Reduced Chi^2 = 1.3412      Levenberg-Marquardt = 0.01
 Free parameter values:
         9.804         0.6682           23.9         0.3721
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 5.0766s    Reduced Chi^2 = 1.3442      Levenberg-Marquardt = 0.001
 Free parameter values:
         10.02         0.6692          23.88         0.3717
 Total time = 5.3734s    Reduced Chi^2 = 1.3562      Levenberg-Marquardt = 0.01
 Free parameter values:
         10.01         0.6691          23.88         0.3717
 Total time = 5.6505s    Reduced Chi^2 = 1.3618      Levenberg-Marquardt = 0.1
 Free parameter values:
         9.982         0.6684          23.88         0.3718
 Total time = 5.9381s    Reduced Chi^2 = 1.3596      Levenberg-Marquardt = 10
 Free parameter values:
         9.853         0.6678          23.89         0.3721
 Total time = 6.2302s    Reduced Chi^2 = 1.3259      Levenberg-Marquardt = 1000
 Free parameter values:
         9.809         0.6682           23.9         0.3721
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 7.3323s    Reduced Chi^2 = 1.3497      Levenberg-Marquardt = 100
 Free parameter values:
          9.81          0.668           23.9         0.3721
 Total time = 7.6062s    Reduced Chi^2 = 1.3614      Levenberg-Marquardt = 1000
 Free parameter values:
         9.809         0.6681           23.9         0.3721
 Total time = 7.8863s    Reduced Chi^2 = 1.3359      Levenberg-Marquardt = 10000
 Free parameter values:
         9.809         0.6682           23.9         0.3721
 Total time = 8.1774s    Reduced Chi^2 = 1.3518      Levenberg-Marquardt = 1000000
 Free parameter values:
         9.809         0.6682           23.9         0.3721
 Total time = 8.4741s    Reduced Chi^2 = 1.3494      Levenberg-Marquardt = 100000000
 Free parameter values:
         9.809         0.6682           23.9         0.3721
 Total time = 8.7633s    Reduced Chi^2 = 1.3485      Levenberg-Marquardt = 10000000000
 Free parameter values:
         9.809         0.6682           23.9         0.3721
 *** No improvement in chi-squared over previous iteration ***
 Total time = 8.7718s    Reduced Chi^2 = 1.3259
 Free parameter values:
         9.809         0.6682           23.9         0.3721
-------------------------------------------------------------------------------
Fit converged:

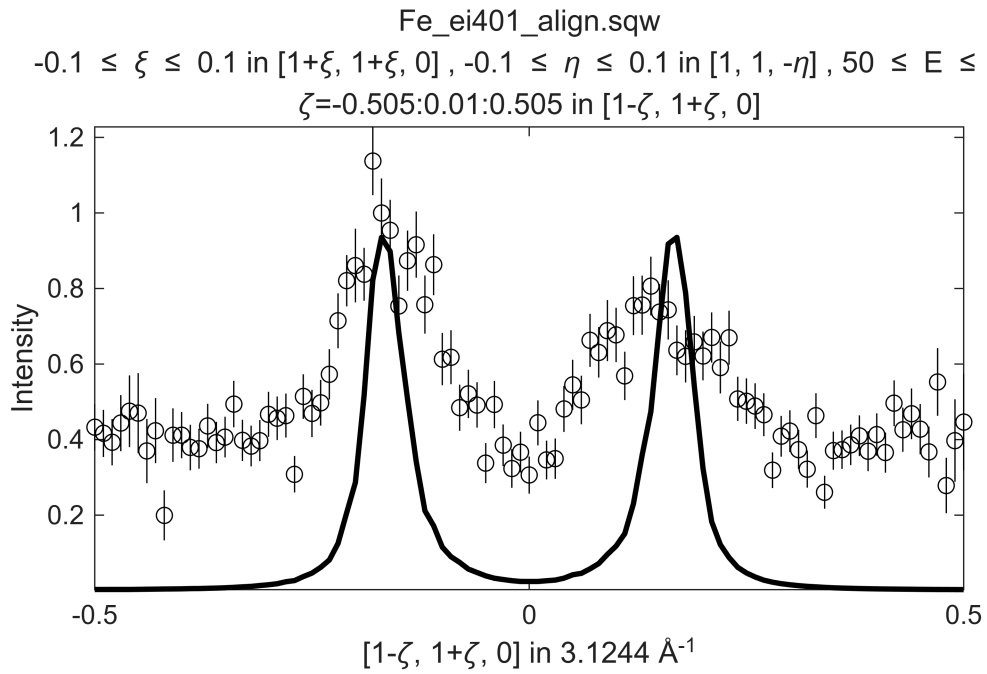
Parameter values (free parameters with error estimates):
---------------------------------------------------------
```

```
Foreground parameter values:
---------------------------
    1              1
    2              8
    3          9.809   +/-   0.4217
    4         0.6682   +/-   0.0282
    5              0
    6           23.9   +/-   0.5614
    7              0
    8              0
    9              0
   10              0


Background parameter values:
---------------------------
    1         0.3721   +/-   0.009074
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000   -0.6324   -0.6315   -0.0598
   -0.6324    1.0000   -0.0081   -0.6082
   -0.6315   -0.0081    1.0000    0.1637
   -0.0598   -0.6082    0.1637    1.0000
```






```
*****************************
**** En = 65±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.29701s    Reduced Chi^2 = 8.3843
 Free parameter values:
         9.809         0.6682           23.9         0.4738
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 1.4079s    Reduced Chi^2 = 1.0622      Levenberg-Marquardt = 0.1
 Free parameter values:
         12.37         0.6973          25.24          0.328
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 2.5366s    Reduced Chi^2 = 1.0203      Levenberg-Marquardt = 0.01
 Free parameter values:
         16.01         0.8019          25.73         0.3088
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 3.6498s    Reduced Chi^2 = 1.0164      Levenberg-Marquardt = 0.001
 Free parameter values:
         15.81         0.8083          25.64         0.3084
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 4.7572s    Reduced Chi^2 = 1.0227      Levenberg-Marquardt = 0.0001
 Free parameter values:
            16         0.8135           25.7         0.3079
 Total time = 5.0541s    Reduced Chi^2 = 1.0231      Levenberg-Marquardt = 0.001
 Free parameter values:
         15.99         0.8134           25.7         0.3079
 Total time = 5.3526s    Reduced Chi^2 = 1.031      Levenberg-Marquardt = 0.01
 Free parameter values:
         15.98         0.8131           25.7          0.308
 Total time = 5.6305s    Reduced Chi^2 = 1.0248      Levenberg-Marquardt = 1
 Free parameter values:
         15.84         0.8097          25.68         0.3083
 Total time = 5.9233s    Reduced Chi^2 = 1.0104      Levenberg-Marquardt = 100
 Free parameter values:
```

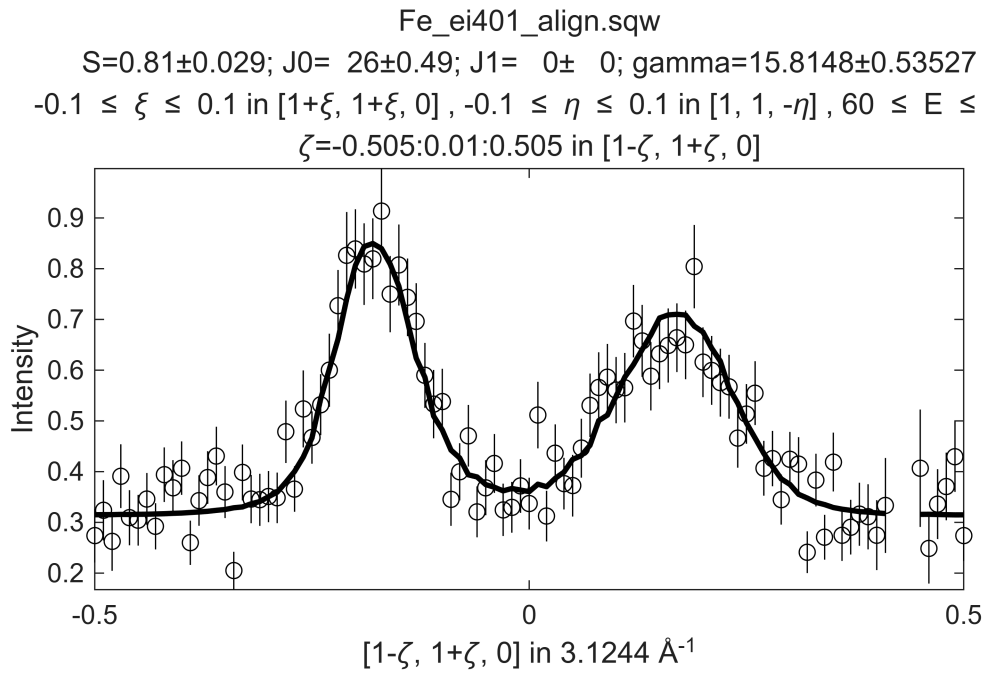
```
        15.81         0.8085          25.65         0.3084
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 7.0614s    Reduced Chi^2 = 1.0256      Levenberg-Marquardt = 10
 Free parameter values:
        15.79         0.8085          25.67         0.3084
 Total time = 7.3424s    Reduced Chi^2 = 1.0363      Levenberg-Marquardt = 100
 Free parameter values:
        15.81         0.8085          25.65         0.3084
 Total time = 7.6275s    Reduced Chi^2 = 1.0301      Levenberg-Marquardt = 1000
 Free parameter values:
        15.81         0.8085          25.65         0.3084
 Total time = 7.9291s    Reduced Chi^2 = 1.0138      Levenberg-Marquardt = 100000
 Free parameter values:
        15.81         0.8085          25.65         0.3084
 Total time = 8.2319s    Reduced Chi^2 = 1.0151      Levenberg-Marquardt = 10000000
 Free parameter values:
        15.81         0.8085          25.65         0.3084
 Total time = 8.5483s    Reduced Chi^2 = 1.0218      Levenberg-Marquardt = 1000000000
 Free parameter values:
        15.81         0.8085          25.65         0.3084
 *** No improvement in chi-squared over previous iteration ***
 Total time = 8.5554s    Reduced Chi^2 = 1.0104
 Free parameter values:
        15.81         0.8085          25.65         0.3084
-------------------------------------------------------------------------------
Fit converged:

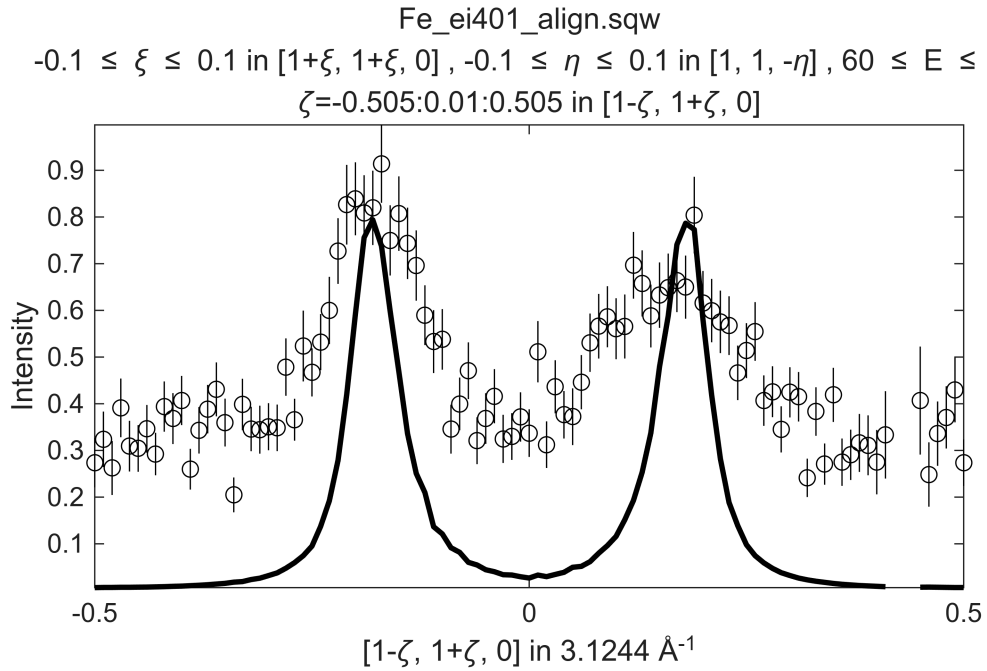
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          15.81   +/-   0.5353
    4         0.8085   +/-   0.02927
    5              0
    6          25.65   +/-   0.4932
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.3084   +/-   0.007856
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000   -0.5024   -0.6129   -0.1507
   -0.5024    1.0000   -0.1061   -0.6742
   -0.6129   -0.1061    1.0000    0.2096
   -0.1507   -0.6742    0.2096    1.0000
```






```
*****************************
**** En = 75±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------------
Starting point
-----------------
 Total time = 0.31836s    Reduced Chi^2 = 9.364
 Free parameter values:
         15.81         0.8085          25.65         0.3973
-------------------------------------------------------------------------------
Iteration = 1
-----------------
 Total time = 1.5021s    Reduced Chi^2 = 0.75397      Levenberg-Marquardt = 0.1
 Free parameter values:
         23.48         0.8364           26.2         0.2555
-------------------------------------------------------------------------------
Iteration = 2
-----------------
 Total time = 2.6671s    Reduced Chi^2 = 0.67786      Levenberg-Marquardt = 0.01
 Free parameter values:
```

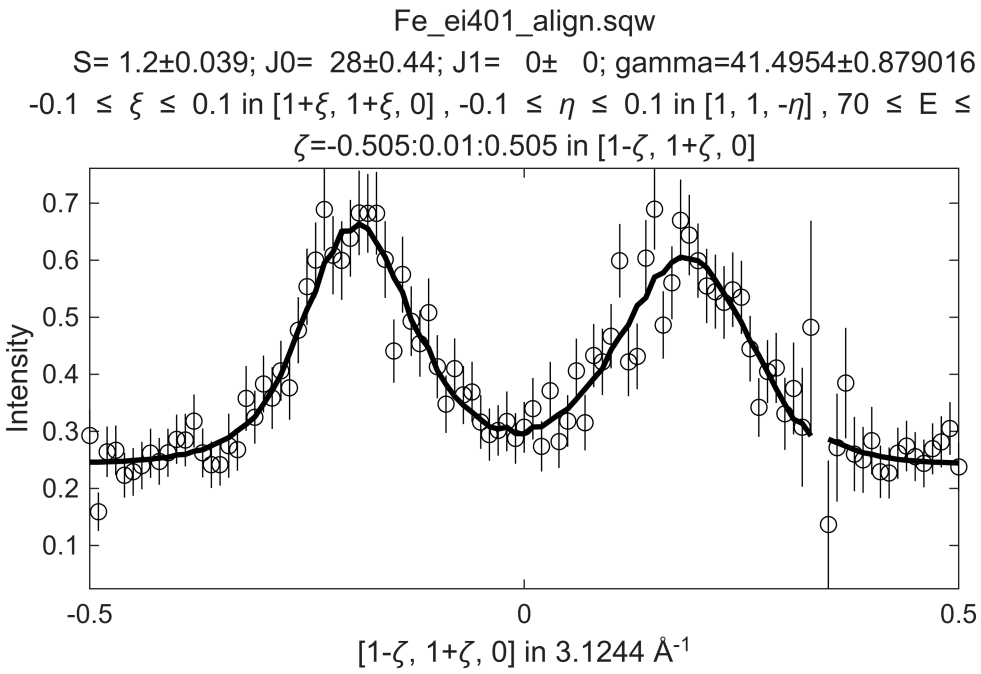
```
        35.19          1.075          27.34         0.2293
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 3.8553s    Reduced Chi^2 = 0.6637      Levenberg-Marquardt = 0.001
 Free parameter values:
        40.13          1.191           28.3         0.2248
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 5.0523s    Reduced Chi^2 = 0.66517      Levenberg-Marquardt = 0.0001
 Free parameter values:
        41.53          1.222          28.48         0.2237
 Total time = 5.3557s    Reduced Chi^2 = 0.659      Levenberg-Marquardt = 0.001
 Free parameter values:
         41.5          1.221          28.47         0.2238
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 6.5261s    Reduced Chi^2 = 0.65923      Levenberg-Marquardt = 0.0001
 Free parameter values:
        41.15          1.216          28.42         0.2236
 Total time = 6.8167s    Reduced Chi^2 = 0.6681      Levenberg-Marquardt = 0.001
 Free parameter values:
        41.15          1.216          28.42         0.2236
 Total time = 7.1215s    Reduced Chi^2 = 0.66661      Levenberg-Marquardt = 0.01
 Free parameter values:
         41.2          1.217          28.43         0.2236
 Total time = 7.423s    Reduced Chi^2 = 0.66068      Levenberg-Marquardt = 1
 Free parameter values:
        41.41          1.221          28.46         0.2236
 Total time = 7.7158s    Reduced Chi^2 = 0.67084      Levenberg-Marquardt = 100
 Free parameter values:
        41.49          1.221          28.47         0.2237
 Total time = 8.0072s    Reduced Chi^2 = 0.65935      Levenberg-Marquardt = 10000
 Free parameter values:
        41.49          1.221          28.47         0.2238
 *** No improvement in chi-squared over previous iteration ***
 Total time = 8.0142s    Reduced Chi^2 = 0.659
 Free parameter values:
         41.5          1.221          28.47         0.2238
-------------------------------------------------------------------------------
Fit converged:

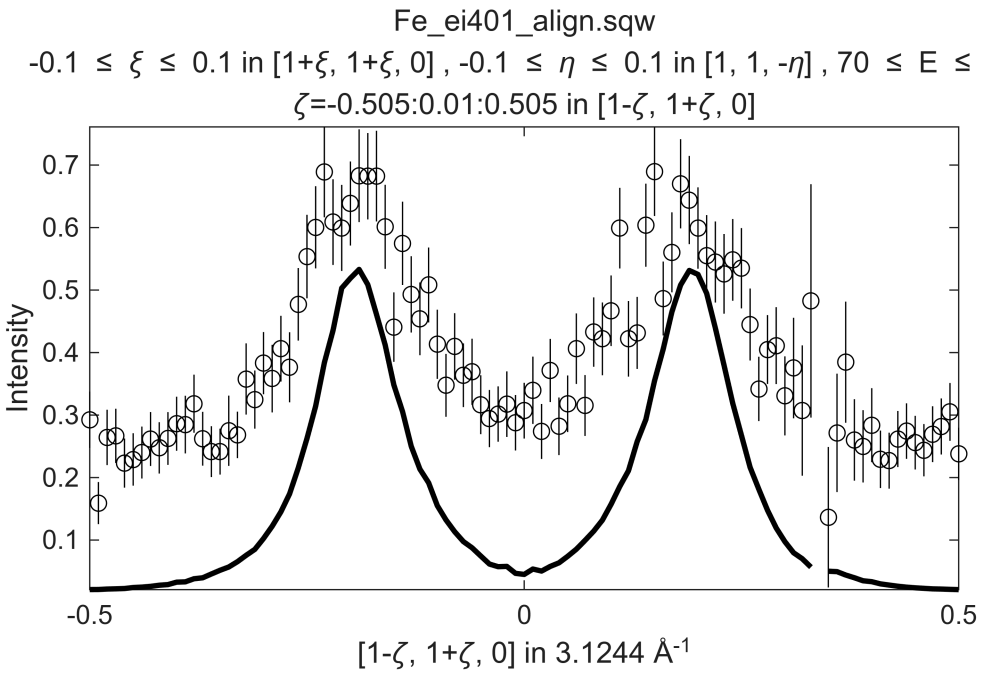
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3           41.5   +/-   0.879
    4          1.221   +/-   0.03913
    5              0
    6          28.47   +/-   0.4372
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.2238   +/-   0.006655
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
    1.0000   -0.0746   -0.5724   -0.3376
   -0.0746    1.0000   -0.3606   -0.8385
   -0.5724   -0.3606    1.0000    0.2806
   -0.3376   -0.8385    0.2806    1.0000
```






```
*****************************
**** En = 85±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.32542s    Reduced Chi^2 = 14.5426
 Free parameter values:
          41.5          1.221          28.47         0.3229
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 1.6454s    Reduced Chi^2 = 0.63082      Levenberg-Marquardt = 0.1
 Free parameter values:
```

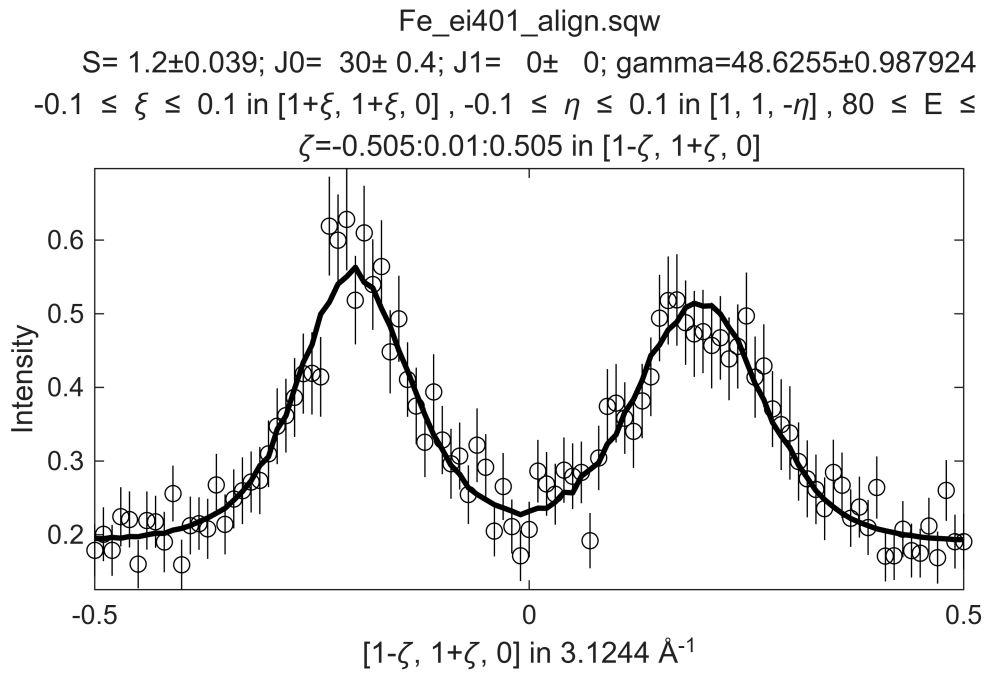
```
        39.86          1.011          29.05          0.189
------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.0936s    Reduced Chi^2 = 0.60148      Levenberg-Marquardt = 0.01
 Free parameter values:
        45.49          1.154          29.52         0.1719
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 4.5569s    Reduced Chi^2 = 0.59572      Levenberg-Marquardt = 0.001
 Free parameter values:
        48.17          1.207          29.82          0.169
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 5.9763s    Reduced Chi^2 = 0.59671      Levenberg-Marquardt = 0.0001
 Free parameter values:
        48.76          1.218           29.9         0.1686
 Total time = 6.333s    Reduced Chi^2 = 0.59947      Levenberg-Marquardt = 0.001
 Free parameter values:
        48.74          1.217           29.9         0.1686
 Total time = 6.7167s    Reduced Chi^2 = 0.59387      Levenberg-Marquardt = 0.01
 Free parameter values:
        48.61          1.215          29.89         0.1687
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 8.1526s    Reduced Chi^2 = 0.60081      Levenberg-Marquardt = 0.001
 Free parameter values:
        49.27          1.228          29.99          0.168
 Total time = 8.535s    Reduced Chi^2 = 0.6014      Levenberg-Marquardt = 0.01
 Free parameter values:
        49.12          1.225          29.97         0.1682
 Total time = 8.8915s    Reduced Chi^2 = 0.60194      Levenberg-Marquardt = 0.1
 Free parameter values:
        48.82          1.219          29.93         0.1685
 Total time = 9.2384s    Reduced Chi^2 = 0.59214      Levenberg-Marquardt = 10
 Free parameter values:
        48.63          1.215          29.89         0.1687
------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 10.7023s    Reduced Chi^2 = 0.59947      Levenberg-Marquardt = 1
 Free parameter values:
        48.69          1.217           29.9         0.1687
 Total time = 11.1073s    Reduced Chi^2 = 0.59528      Levenberg-Marquardt = 10
 Free parameter values:
        48.65          1.216           29.9         0.1687
 Total time = 11.4668s    Reduced Chi^2 = 0.60425      Levenberg-Marquardt = 100
 Free parameter values:
        48.63          1.216          29.89         0.1687
 Total time = 11.8424s    Reduced Chi^2 = 0.60741      Levenberg-Marquardt = 10000
 Free parameter values:
        48.63          1.215          29.89         0.1687
 Total time = 12.2238s    Reduced Chi^2 = 0.59867      Levenberg-Marquardt = 1000000
 Free parameter values:
        48.63          1.215          29.89         0.1687
 Total time = 12.6169s    Reduced Chi^2 = 0.60245      Levenberg-Marquardt = 100000000
 Free parameter values:
        48.63          1.215          29.89         0.1687
 *** No improvement in chi-squared over previous iteration ***
 Total time = 12.6256s    Reduced Chi^2 = 0.59214
 Free parameter values:
        48.63          1.215          29.89         0.1687
------------------------------------------------------------------------------
Fit converged:
```

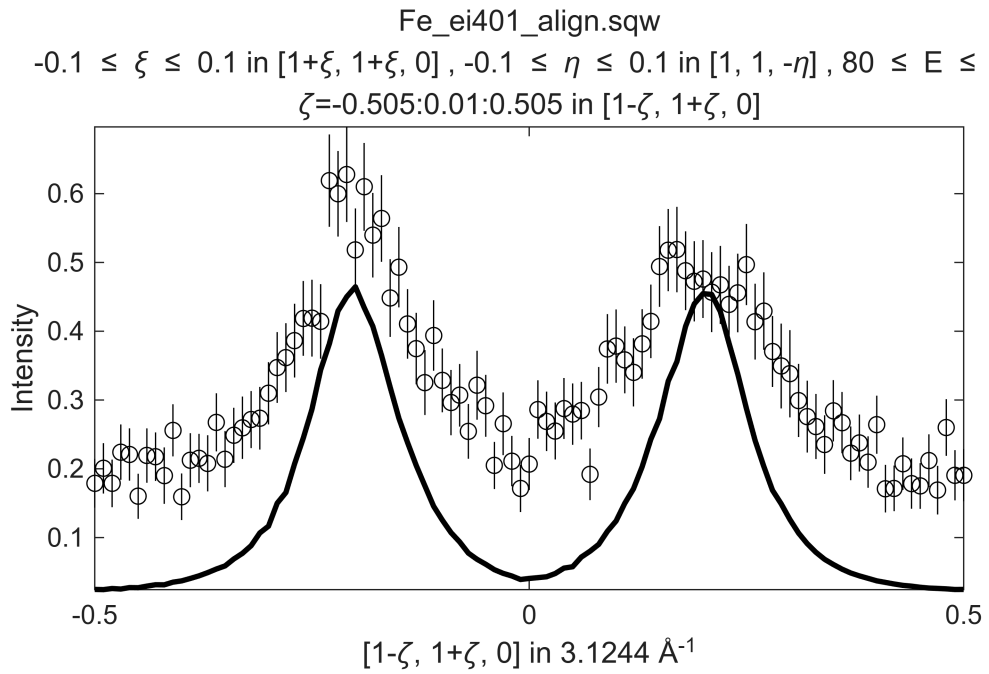
```
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          48.63   +/-   0.9879
    4          1.215   +/-   0.03893
    5              0
    6          29.89   +/-   0.3967
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.1687   +/-   0.00565
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.0960   -0.5635   -0.4740
    0.0960    1.0000   -0.3678   -0.8685
   -0.5635   -0.3678    1.0000    0.2965
   -0.4740   -0.8685    0.2965    1.0000
```






```
*****************************
**** En = 95±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.41081s    Reduced Chi^2 = 17.3351
 Free parameter values:
         48.63          1.215          29.89          0.279
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 1.9216s    Reduced Chi^2 = 0.78477      Levenberg-Marquardt = 0.1
 Free parameter values:
         33.43         0.8873          30.24         0.1558
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.5085s    Reduced Chi^2 = 0.73324      Levenberg-Marquardt = 0.01
 Free parameter values:
         37.03           1.01          30.26           0.14
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.0188s    Reduced Chi^2 = 0.72487      Levenberg-Marquardt = 0.001
 Free parameter values:
         36.84          1.015           30.3         0.1396
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 6.6538s    Reduced Chi^2 = 0.73643      Levenberg-Marquardt = 0.0001
 Free parameter values:
         36.57          1.012           30.3         0.1398
 Total time = 7.0375s    Reduced Chi^2 = 0.73109      Levenberg-Marquardt = 0.001
 Free parameter values:
         36.57          1.012           30.3         0.1398
 Total time = 7.4097s    Reduced Chi^2 = 0.73782      Levenberg-Marquardt = 0.01
 Free parameter values:
         36.59          1.012           30.3         0.1398
 Total time = 7.8036s    Reduced Chi^2 = 0.72979      Levenberg-Marquardt = 1
 Free parameter values:
         36.74          1.014          30.31         0.1397
 Total time = 8.1983s    Reduced Chi^2 = 0.73326      Levenberg-Marquardt = 100
 Free parameter values:
```

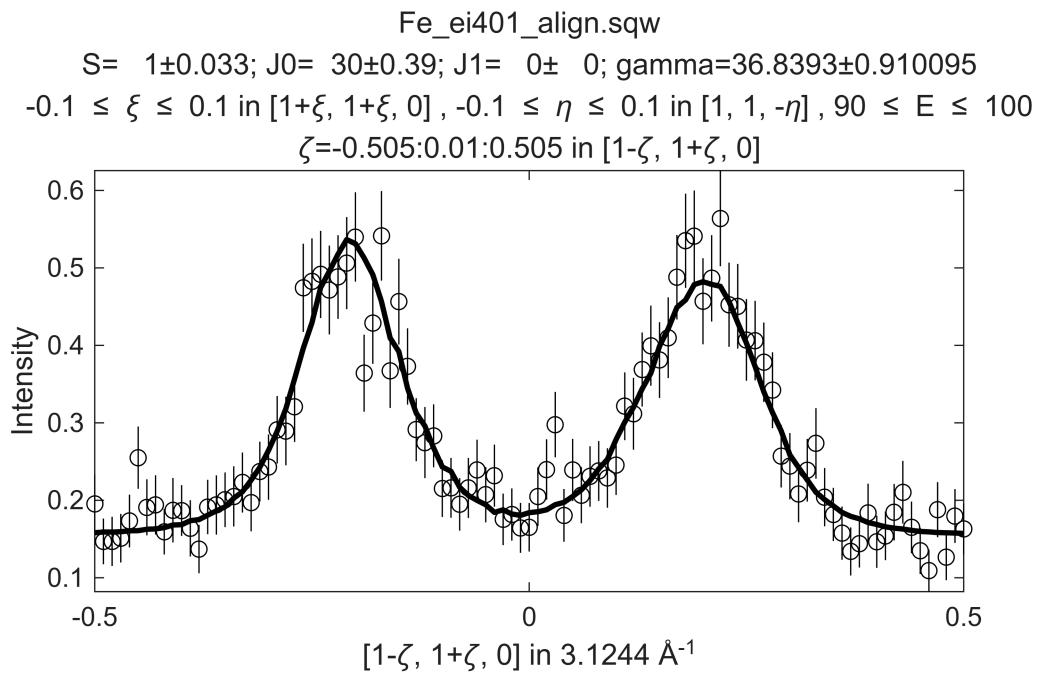
```
        36.83          1.015           30.3         0.1396
 Total time = 8.6005s    Reduced Chi^2 = 0.74078      Levenberg-Marquardt = 10000
 Free parameter values:
        36.84          1.015           30.3         0.1396
 *** No improvement in chi-squared over previous iteration ***
 Total time = 8.609s    Reduced Chi^2 = 0.72487
 Free parameter values:
        36.84          1.015           30.3         0.1396
--------------------------------------------------------------------------------
Fit converged:

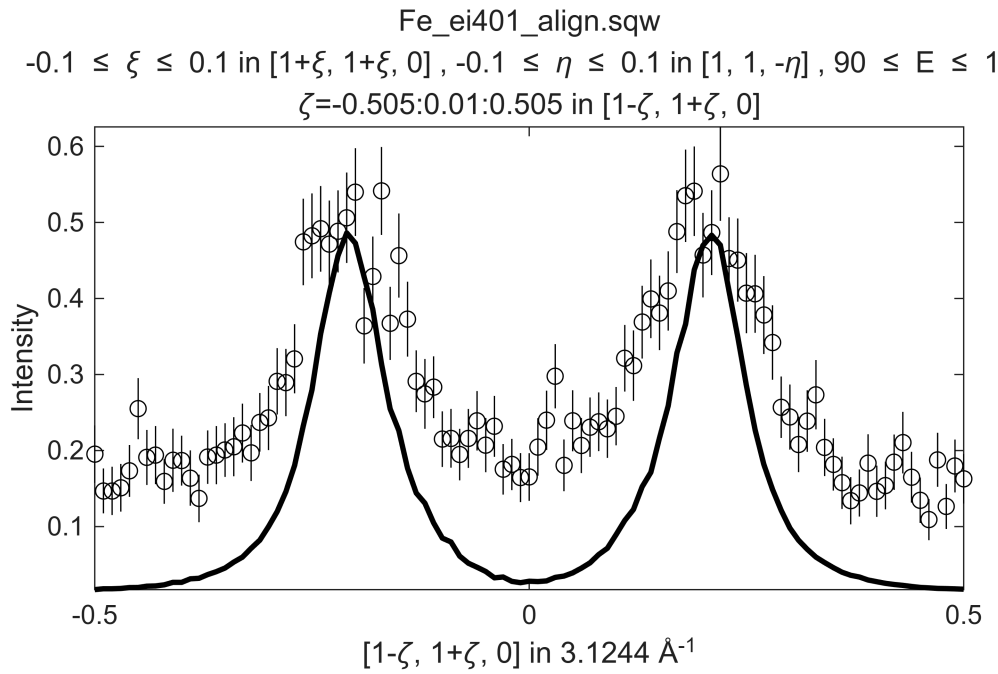
Parameter values (free parameters with error estimates):
------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          36.84   +/-   0.9101
    4          1.015   +/-   0.0332
    5              0
    6           30.3   +/-   0.3896
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.1396   +/-   0.005244
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.0777   -0.5268   -0.5437
    0.0777    1.0000   -0.2684   -0.8309
   -0.5268   -0.2684    1.0000    0.2375
   -0.5437   -0.8309    0.2375    1.0000
```






```
*****************************
**** En = 105±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.46682s    Reduced Chi^2 = 18.7952
 Free parameter values:
         36.84          1.015           30.3         0.2308
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.2719s    Reduced Chi^2 = 0.98168      Levenberg-Marquardt = 0.1
 Free parameter values:
         36.86         0.9176          30.94         0.1084
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.9648s    Reduced Chi^2 = 0.90554      Levenberg-Marquardt = 0.01
 Free parameter values:
         48.08          1.131          31.81        0.08719
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.7544s    Reduced Chi^2 = 0.89764      Levenberg-Marquardt = 0.001
 Free parameter values:
         50.95          1.193          32.22        0.08432
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 7.5911s    Reduced Chi^2 = 0.8917      Levenberg-Marquardt = 0.0001
 Free parameter values:
          51.4            1.2          32.25        0.08395
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 9.4981s    Reduced Chi^2 = 0.89026      Levenberg-Marquardt = 1e-05
 Free parameter values:
         52.13          1.213          32.33        0.08338
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 11.2444s    Reduced Chi^2 = 0.89407      Levenberg-Marquardt = 1e-06
 Free parameter values:
```

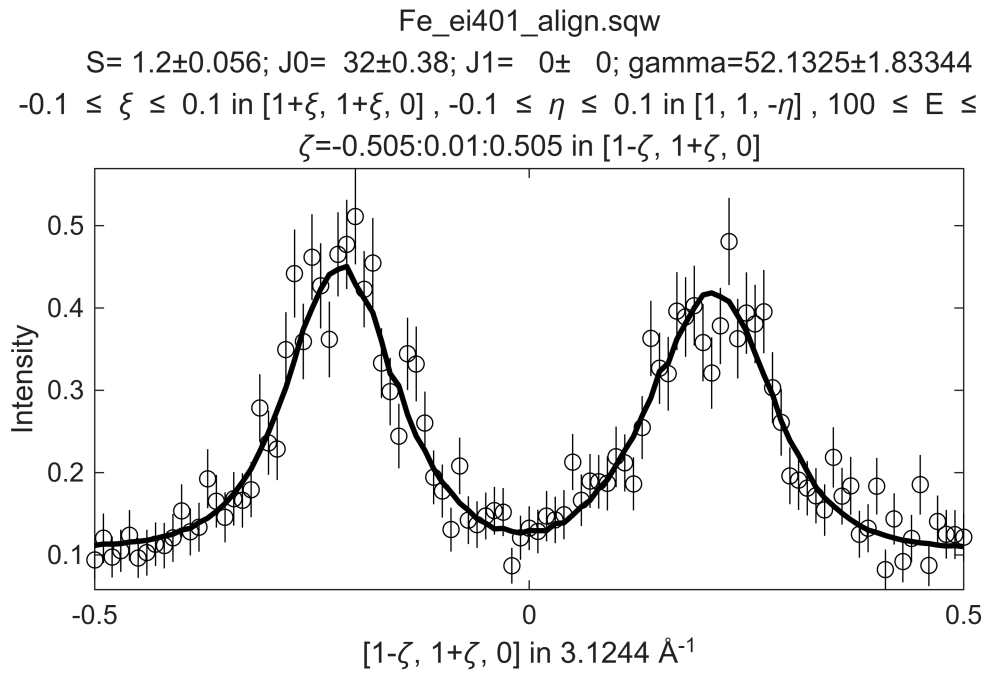
```
         51.6          1.205          32.29        0.08374
 Total time = 11.6969s    Reduced Chi^2 = 0.89684      Levenberg-Marquardt = 1e-05
 Free parameter values:
         51.6          1.205          32.29        0.08374
 Total time = 12.1399s    Reduced Chi^2 = 0.89639      Levenberg-Marquardt = 0.0001
 Free parameter values:
         51.6          1.205          32.29        0.08374
 Total time = 12.5742s    Reduced Chi^2 = 0.89422      Levenberg-Marquardt = 0.01
 Free parameter values:
         51.7          1.207           32.3        0.08365
 Total time = 13.0366s    Reduced Chi^2 = 0.90584      Levenberg-Marquardt = 1
 Free parameter values:
        52.03          1.212          32.33        0.08338
 Total time = 13.445s    Reduced Chi^2 = 0.9014      Levenberg-Marquardt = 100
 Free parameter values:
        52.12          1.213          32.33        0.08338
 *** No improvement in chi-squared over previous iteration ***
 Total time = 13.4521s    Reduced Chi^2 = 0.89026
 Free parameter values:
        52.13          1.213          32.33        0.08338
--------------------------------------------------------------------------------
Fit converged:

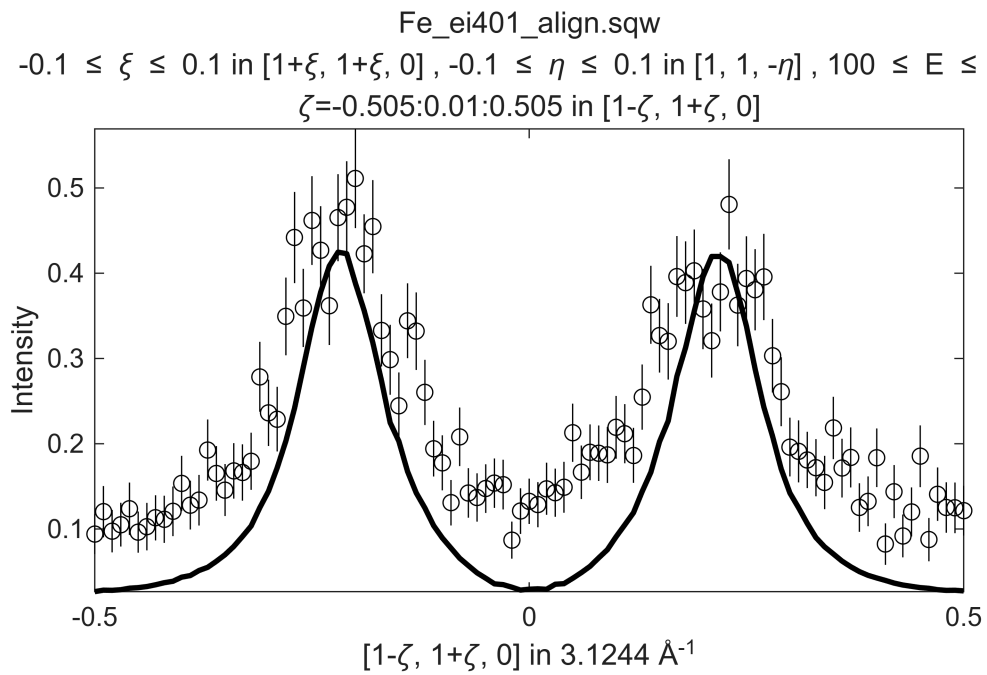
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          52.13   +/-   1.833
    4          1.213   +/-   0.05607
    5              0
    6          32.33   +/-   0.3806
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.08338   +/-   0.006099
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.6448   -0.2210   -0.8524
    0.6448    1.0000   -0.0662   -0.9310
   -0.2210   -0.0662    1.0000   -0.0133
   -0.8524   -0.9310   -0.0133    1.0000
```






```
*****************************
**** En = 115±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.44487s    Reduced Chi^2 = 25.1381
 Free parameter values:
         52.13          1.213          32.33         0.1975
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.1623s    Reduced Chi^2 = 0.88525      Levenberg-Marquardt = 0.1
 Free parameter values:
         41.25         0.8291          32.59        0.09094
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.9084s    Reduced Chi^2 = 0.80617      Levenberg-Marquardt = 0.01
 Free parameter values:
```

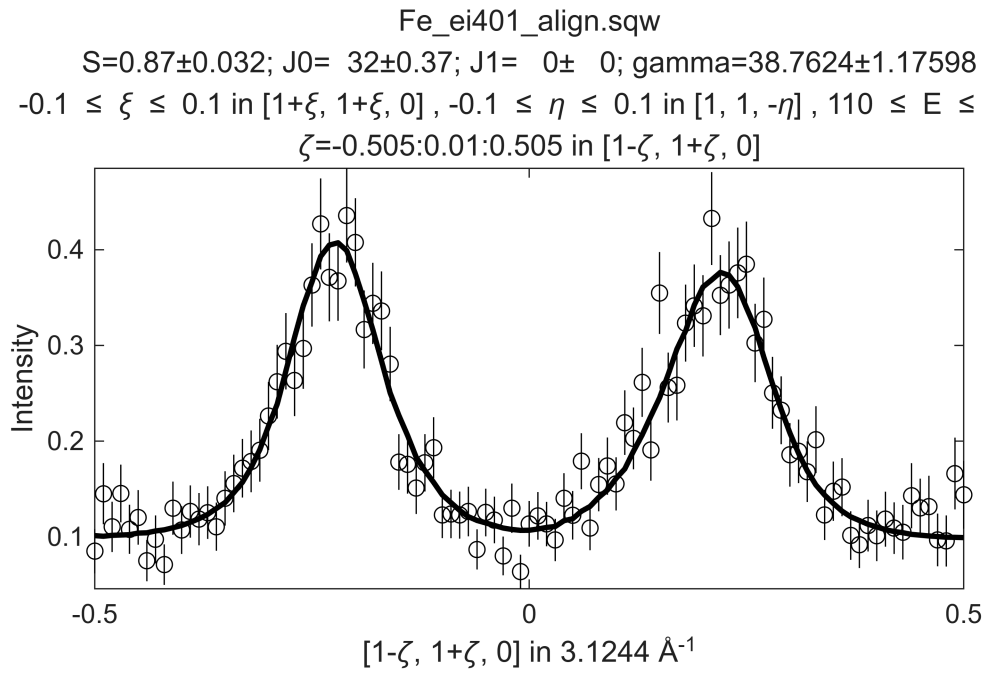
```
        38.73         0.8685          32.43        0.08264
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.6369s    Reduced Chi^2 = 0.81175      Levenberg-Marquardt = 0.001
 Free parameter values:
        39.07          0.872          32.51        0.08253
 Total time = 6.0657s    Reduced Chi^2 = 0.808      Levenberg-Marquardt = 0.01
 Free parameter values:
        39.05         0.8716          32.51        0.08256
 Total time = 6.4913s    Reduced Chi^2 = 0.81456      Levenberg-Marquardt = 0.1
 Free parameter values:
        38.94         0.8701           32.5        0.08266
 Total time = 6.9073s    Reduced Chi^2 = 0.80159      Levenberg-Marquardt = 10
 Free parameter values:
        38.76         0.8686          32.45        0.08266
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 8.6144s    Reduced Chi^2 = 0.80542      Levenberg-Marquardt = 1
 Free parameter values:
        38.86         0.8696          32.47        0.08269
 Total time = 9.0684s    Reduced Chi^2 = 0.80568      Levenberg-Marquardt = 10
 Free parameter values:
        38.79         0.8689          32.46        0.08267
 Total time = 9.5065s    Reduced Chi^2 = 0.82031      Levenberg-Marquardt = 100
 Free parameter values:
        38.77         0.8687          32.45        0.08266
 Total time = 9.9409s    Reduced Chi^2 = 0.80532      Levenberg-Marquardt = 10000
 Free parameter values:
        38.76         0.8686          32.45        0.08266
 Total time = 10.376s    Reduced Chi^2 = 0.80512      Levenberg-Marquardt = 1000000
 Free parameter values:
        38.76         0.8686          32.45        0.08266
 Total time = 10.8069s    Reduced Chi^2 = 0.80594      Levenberg-Marquardt = 100000000
 Free parameter values:
        38.76         0.8686          32.45        0.08266
 *** No improvement in chi-squared over previous iteration ***
 Total time = 10.8163s    Reduced Chi^2 = 0.80159
 Free parameter values:
        38.76         0.8686          32.45        0.08266
-------------------------------------------------------------------------------
Fit converged:

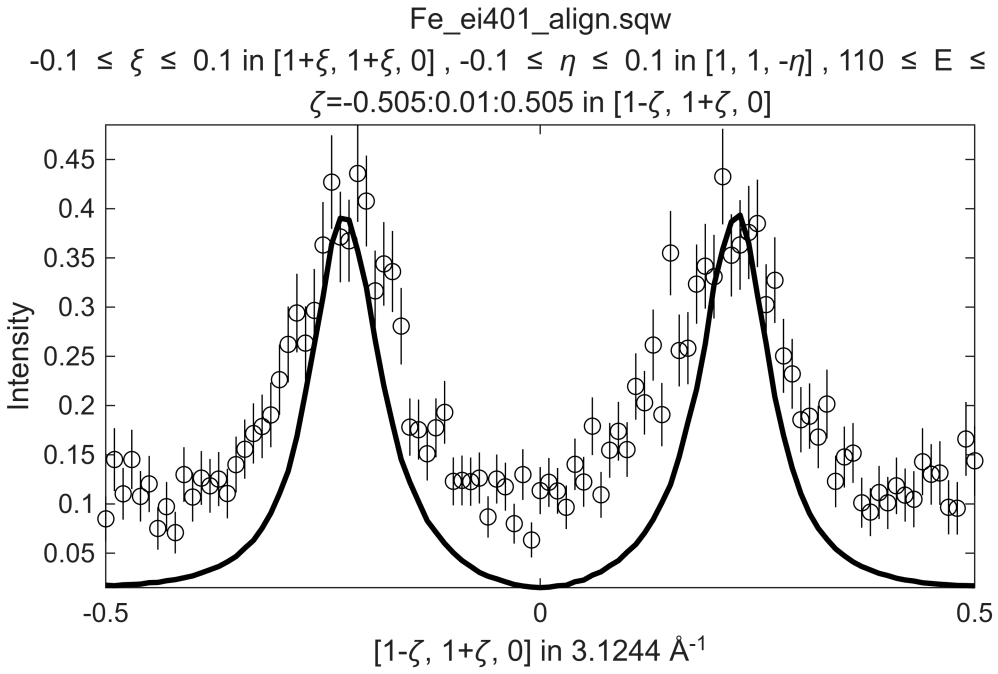
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          38.76   +/-   1.176
    4         0.8686   +/-   0.03226
    5              0
    6          32.45   +/-   0.366
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.08266   +/-   0.00444
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.3636   -0.3573   -0.7407
    0.3636    1.0000   -0.1382   -0.8680
   -0.3573   -0.1382    1.0000    0.0876
   -0.7407   -0.8680    0.0876    1.0000
```






```
*****************************
**** En = 125±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.46831s    Reduced Chi^2 = 22.5499
 Free parameter values:
         38.76         0.8686          32.45         0.1735
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.3403s    Reduced Chi^2 = 0.89992      Levenberg-Marquardt = 0.1
 Free parameter values:
```

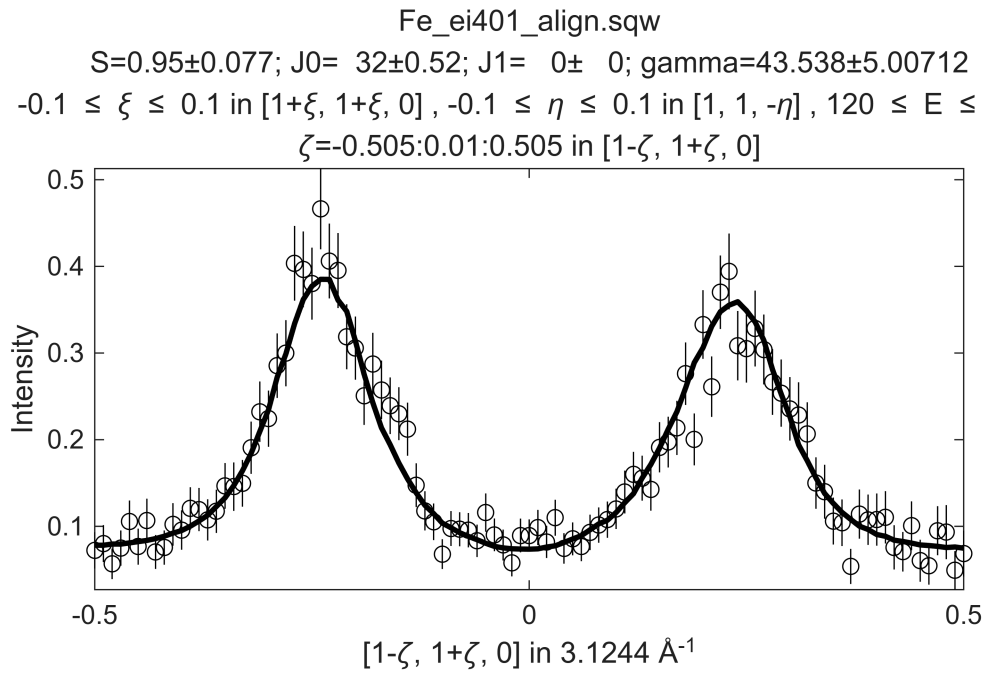
```
         34.31         0.7832          31.73        0.06899
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 4.2139s    Reduced Chi^2 = 0.81275      Levenberg-Marquardt = 0.01
 Free parameter values:
         42.61           0.93           32.1        0.05285
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 6.0429s    Reduced Chi^2 = 0.81052      Levenberg-Marquardt = 0.001
 Free parameter values:
         43.45         0.9517          32.31        0.05176
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 7.8843s    Reduced Chi^2 = 0.81267      Levenberg-Marquardt = 0.0001
 Free parameter values:
         43.76         0.9565          32.32        0.05149
 Total time = 8.354s    Reduced Chi^2 = 0.80637      Levenberg-Marquardt = 0.001
 Free parameter values:
         43.75         0.9564          32.32         0.0515
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 10.147s    Reduced Chi^2 = 0.80562      Levenberg-Marquardt = 0.0001
 Free parameter values:
         43.54         0.9515          32.29        0.05167
-------------------------------------------------------------------------------
Fit converged:

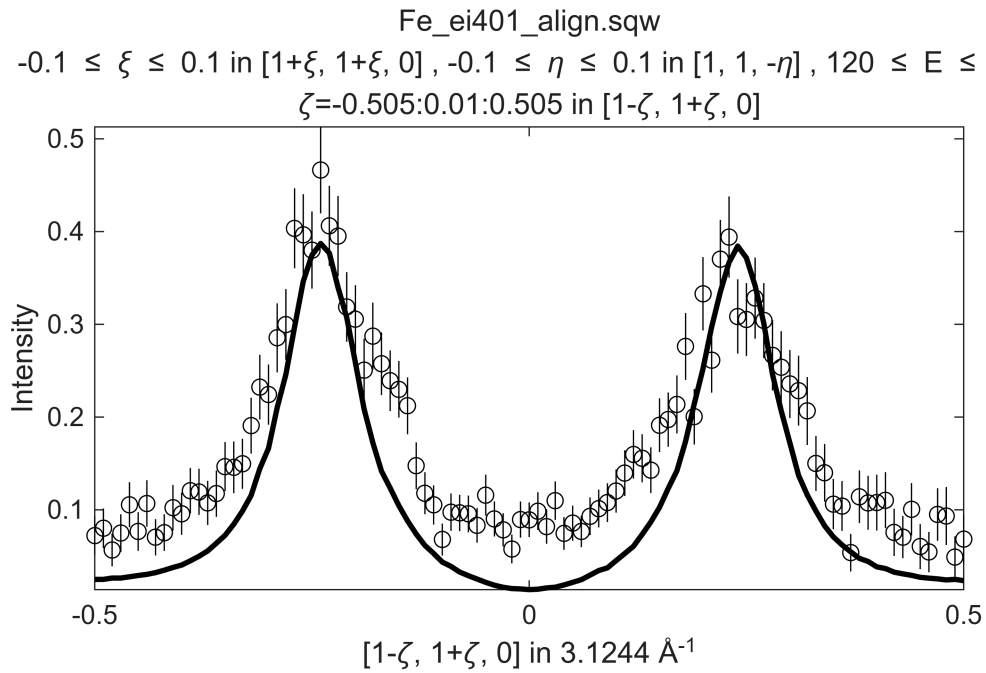
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          43.54   +/-   5.007
    4         0.9515   +/-   0.0772
    5              0
    6          32.29   +/-   0.5234
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.05167   +/-   0.005735
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9267    0.7687   -0.8290
    0.9267    1.0000    0.7240   -0.9135
    0.7687    0.7240    1.0000   -0.5870
   -0.8290   -0.9135   -0.5870    1.0000
```






```
*****************************
**** En = 135±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.52267s    Reduced Chi^2 = 31.8372
 Free parameter values:
         43.54         0.9515          32.29         0.1553
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.5301s    Reduced Chi^2 = 1.2836      Levenberg-Marquardt = 0.1
 Free parameter values:
         42.21         0.8517          33.42        0.05167
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 4.5915s    Reduced Chi^2 = 1.1366      Levenberg-Marquardt = 0.01
 Free parameter values:
```

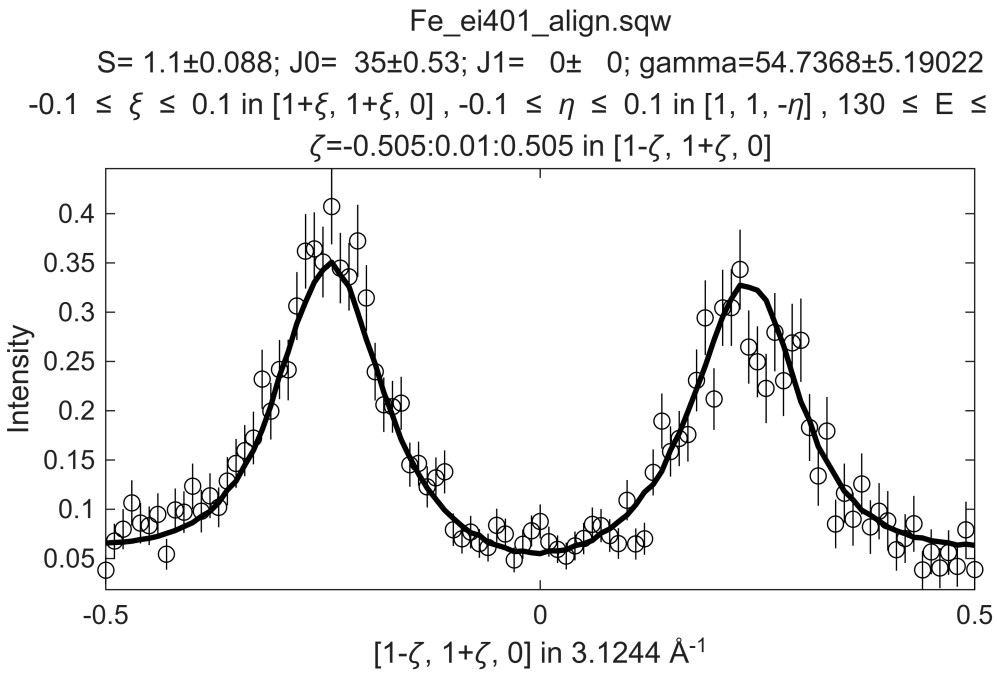
```
        53.08          1.038          34.37        0.03407
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 6.5738s    Reduced Chi^2 = 1.1373      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.99          1.078          34.59        0.03234
 Total time = 7.0925s    Reduced Chi^2 = 1.1242      Levenberg-Marquardt = 0.01
 Free parameter values:
        54.74          1.075          34.56        0.03257
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 9.1379s    Reduced Chi^2 = 1.139      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.77          1.077          34.58        0.03248
 Total time = 9.6579s    Reduced Chi^2 = 1.1255      Levenberg-Marquardt = 0.01
 Free parameter values:
        54.75          1.077          34.58        0.03249
 Total time = 10.1821s    Reduced Chi^2 = 1.1299      Levenberg-Marquardt = 0.1
 Free parameter values:
        54.71          1.076          34.57        0.03253
 Total time = 10.7044s    Reduced Chi^2 = 1.1398      Levenberg-Marquardt = 10
 Free parameter values:
        54.71          1.075          34.57        0.03258
 Total time = 11.2188s    Reduced Chi^2 = 1.1308      Levenberg-Marquardt = 1000
 Free parameter values:
        54.73          1.075          34.56        0.03257
 Total time = 11.7343s    Reduced Chi^2 = 1.1245      Levenberg-Marquardt = 100000
 Free parameter values:
        54.74          1.075          34.56        0.03257
 *** No improvement in chi-squared over previous iteration ***
 Total time = 11.7412s    Reduced Chi^2 = 1.1242
 Free parameter values:
        54.74          1.075          34.56        0.03257
-------------------------------------------------------------------------------
Fit converged:

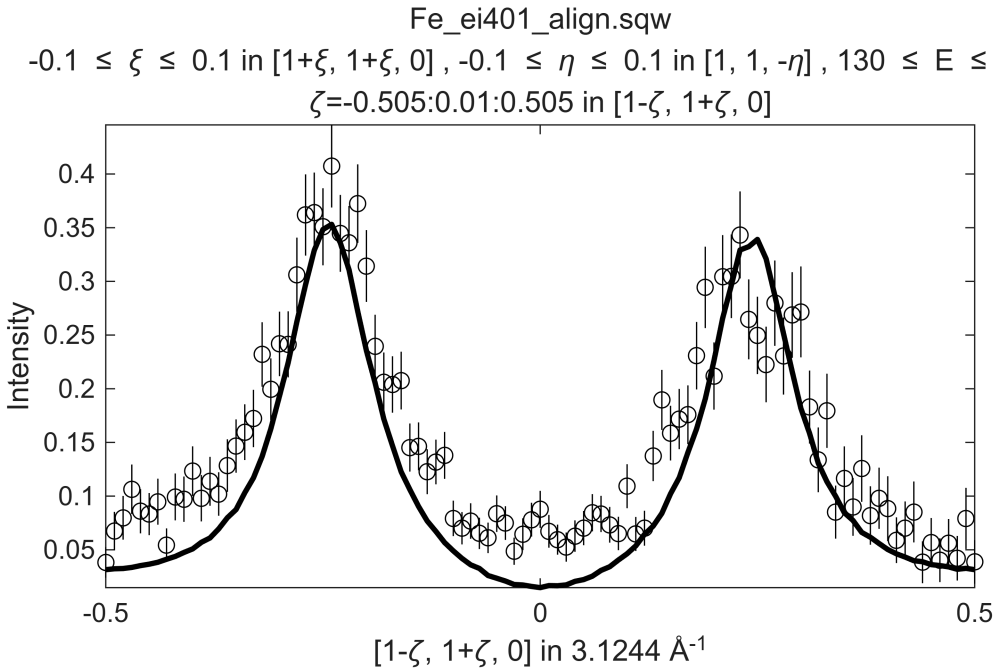
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          54.74   +/-   5.19
    4          1.075   +/-   0.08846
    5              0
    6          34.56   +/-   0.5305
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.03257   +/-   0.006044
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9319    0.7094   -0.9830
    0.9319    1.0000    0.7224   -0.9794
    0.7094    0.7224    1.0000   -0.7657
   -0.9830   -0.9794   -0.7657    1.0000
```






```
*****************************
**** En = 145±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.58797s    Reduced Chi^2 = 40.9893
 Free parameter values:
         54.74          1.075          34.56         0.1437
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.9329s    Reduced Chi^2 = 0.8946      Levenberg-Marquardt = 0.1
 Free parameter values:
         46.44         0.8511           34.7        0.04452
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 5.2928s    Reduced Chi^2 = 0.77982      Levenberg-Marquardt = 0.01
 Free parameter values:
```

```
         51.93         0.9642           35.1        0.03286
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 7.7155s    Reduced Chi^2 = 0.79021      Levenberg-Marquardt = 0.001
 Free parameter values:
         53.02         0.9845          35.19        0.03178
 Total time = 8.3161s    Reduced Chi^2 = 0.78269      Levenberg-Marquardt = 0.01
 Free parameter values:
         52.91          0.983          35.18        0.03187
 Total time = 8.9333s    Reduced Chi^2 = 0.77752      Levenberg-Marquardt = 0.1
 Free parameter values:
         52.48         0.9766          35.14        0.03225
-------------------------------------------------------------------------------
Iteration = 4
-----------------
 Total time = 11.3058s    Reduced Chi^2 = 0.77896      Levenberg-Marquardt = 0.01
 Free parameter values:
         52.93         0.9843          35.19        0.03179
 Total time = 11.8979s    Reduced Chi^2 = 0.77267      Levenberg-Marquardt = 0.1
 Free parameter values:
         52.73         0.9815          35.17        0.03196
-------------------------------------------------------------------------------
Iteration = 5
-----------------
 Total time = 14.2157s    Reduced Chi^2 = 0.77732      Levenberg-Marquardt = 0.01
 Free parameter values:
          52.9         0.9836          35.18        0.03179
 Total time = 14.8295s    Reduced Chi^2 = 0.78108      Levenberg-Marquardt = 0.1
 Free parameter values:
         52.84         0.9827          35.17        0.03184
 Total time = 15.454s    Reduced Chi^2 = 0.78074      Levenberg-Marquardt = 1
 Free parameter values:
         52.78         0.9819          35.17         0.0319
 Total time = 16.0163s    Reduced Chi^2 = 0.78466      Levenberg-Marquardt = 100
 Free parameter values:
         52.74         0.9815          35.17        0.03195
 Total time = 16.6455s    Reduced Chi^2 = 0.77707      Levenberg-Marquardt = 10000
 Free parameter values:
         52.73         0.9815          35.17        0.03195
 Total time = 17.2488s    Reduced Chi^2 = 0.78445      Levenberg-Marquardt = 1000000
 Free parameter values:
         52.73         0.9815          35.17        0.03196
 *** No improvement in chi-squared over previous iteration ***
 Total time = 17.2572s    Reduced Chi^2 = 0.77267
 Free parameter values:
         52.73         0.9815          35.17        0.03196
-------------------------------------------------------------------------------
Fit converged:

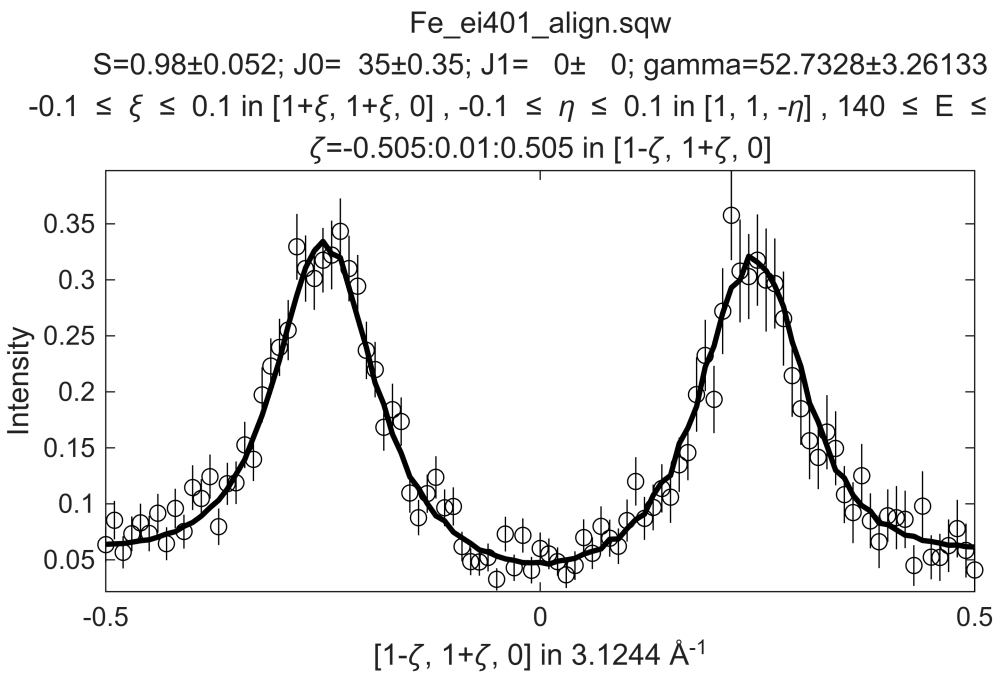
Parameter values (free parameters with error estimates):
---------------------------------------------------------

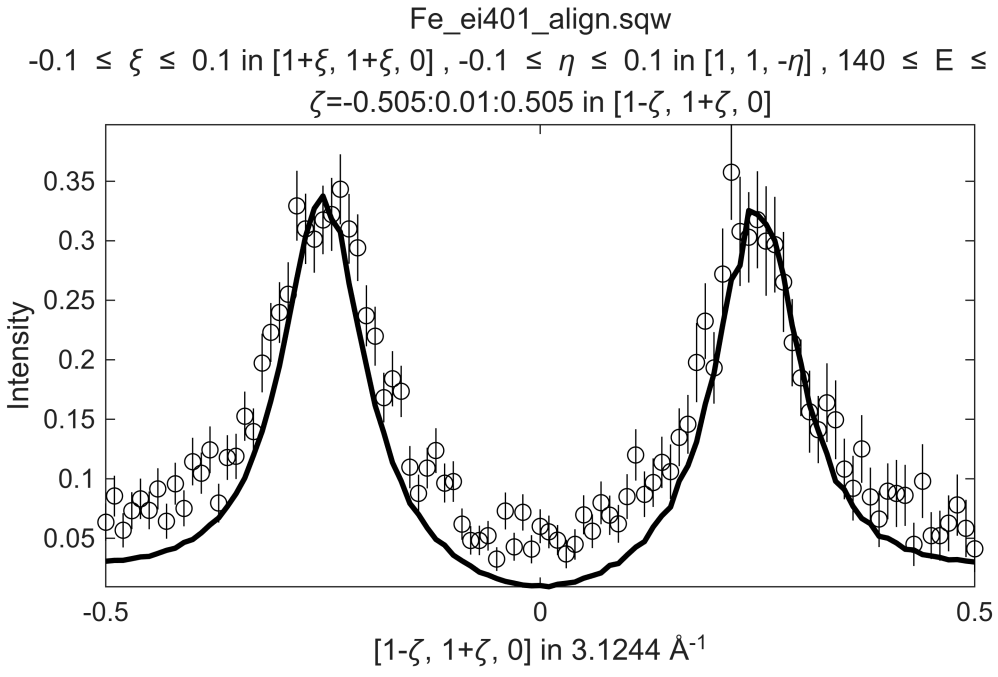
Foreground parameter values:
----------------------------
    1              1
    2              8
    3          52.73   +/-   3.261
    4         0.9815   +/-   0.05237
    5              0
    6          35.17   +/-   0.3461
    7              0
    8              0
    9              0
   10              0


Background parameter values:
```

```
----------------------------
    1        0.03196   +/-   0.003793
    2              0
    3              0


Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.8938    0.6103   -0.9737
    0.8938    1.0000    0.6396   -0.9684
    0.6103    0.6396    1.0000   -0.6887
   -0.9737   -0.9684   -0.6887    1.0000
```






```
*****************************
**** En = 155±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.56517s    Reduced Chi^2 = 37.2682
 Free parameter values:
```

```
         52.73         0.9815          35.17         0.1382
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.8521s    Reduced Chi^2 = 1.0183      Levenberg-Marquardt = 0.1
 Free parameter values:
         44.63         0.8597          34.25        0.03782
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 5.0886s    Reduced Chi^2 = 0.86153      Levenberg-Marquardt = 0.01
 Free parameter values:
         52.98          1.008           34.7        0.02394
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 7.3771s    Reduced Chi^2 = 0.86625      Levenberg-Marquardt = 0.001
 Free parameter values:
         54.23          1.035          34.83         0.0226
 Total time = 7.9846s    Reduced Chi^2 = 0.85691      Levenberg-Marquardt = 0.01
 Free parameter values:
         54.11          1.033          34.82        0.02271
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 10.2348s    Reduced Chi^2 = 0.86317      Levenberg-Marquardt = 0.001
 Free parameter values:
         54.31          1.036          34.83        0.02253
 Total time = 10.8242s    Reduced Chi^2 = 0.86051      Levenberg-Marquardt = 0.01
 Free parameter values:
         54.29          1.036          34.83        0.02254
 Total time = 11.4433s    Reduced Chi^2 = 0.85481      Levenberg-Marquardt = 0.1
 Free parameter values:
         54.22          1.035          34.83         0.0226
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 13.7496s    Reduced Chi^2 = 0.8627      Levenberg-Marquardt = 0.01
 Free parameter values:
         54.28          1.037          34.84        0.02252
 Total time = 14.3226s    Reduced Chi^2 = 0.85928      Levenberg-Marquardt = 0.1
 Free parameter values:
         54.24          1.036          34.84        0.02255
 Total time = 14.926s    Reduced Chi^2 = 0.85908      Levenberg-Marquardt = 1
 Free parameter values:
         54.21          1.036          34.83        0.02258
 Total time = 15.484s    Reduced Chi^2 = 0.86482      Levenberg-Marquardt = 100
 Free parameter values:
         54.22          1.035          34.83         0.0226
 Total time = 16.0587s    Reduced Chi^2 = 0.85665      Levenberg-Marquardt = 10000
 Free parameter values:
         54.22          1.035          34.83         0.0226
 Total time = 16.6356s    Reduced Chi^2 = 0.85821      Levenberg-Marquardt = 1000000
 Free parameter values:
         54.22          1.035          34.83         0.0226
 *** No improvement in chi-squared over previous iteration ***
 Total time = 16.6439s    Reduced Chi^2 = 0.85481
 Free parameter values:
         54.22          1.035          34.83         0.0226
--------------------------------------------------------------------------------
Fit converged:

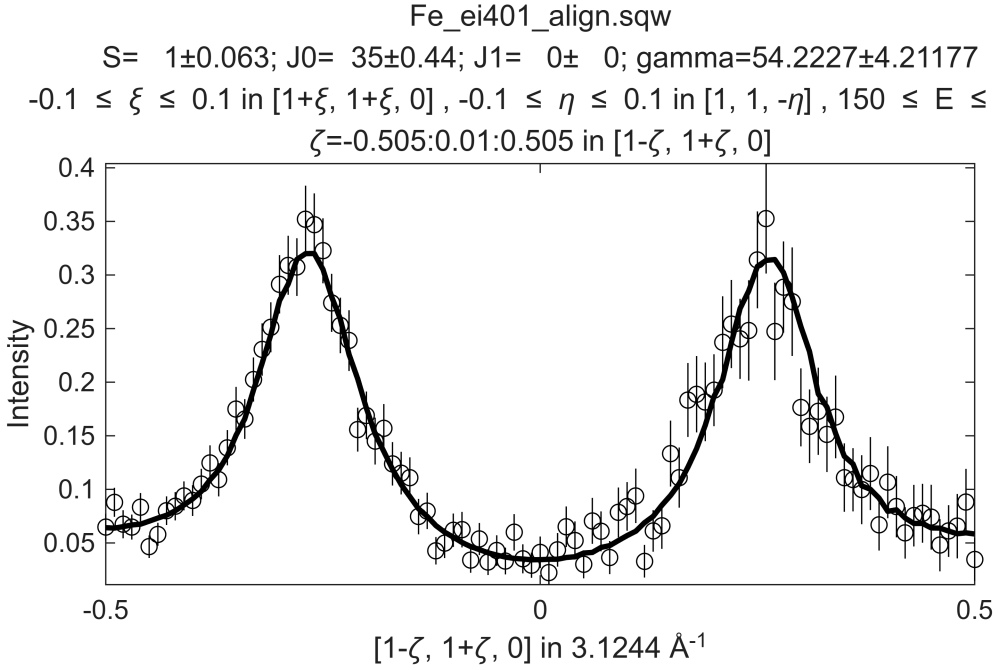
Parameter values (free parameters with error estimates):
---------------------------------------------------------

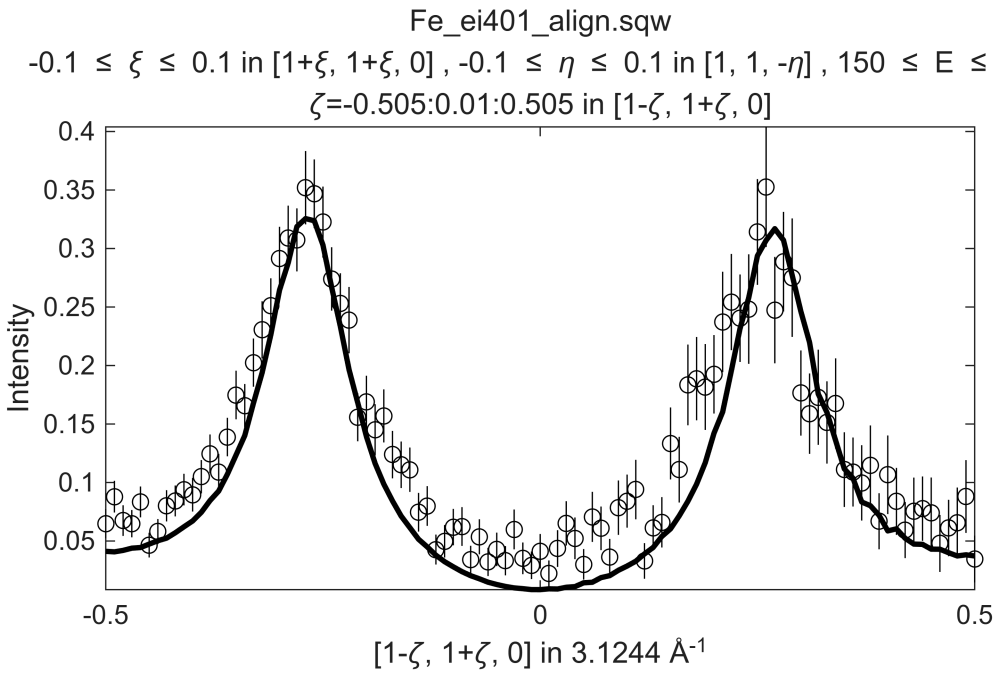
Foreground parameter values:
----------------------------
    1              1
```

```
    2              8
    3          54.22   +/-   4.212
    4          1.035   +/-   0.06265
    5              0
    6          34.83   +/-   0.4432
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.0226   +/-   0.0033
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9105    0.8067   -0.9801
    0.9105    1.0000    0.8329   -0.9721
    0.8067    0.8329    1.0000   -0.8621
   -0.9801   -0.9721   -0.8621    1.0000
```






```
*****************************
**** En = 165±5 going to 165
*****************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.9901
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.39902s    Reduced Chi^2 = 51.6697
 Free parameter values:
         54.22          1.035          34.83         0.1327
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.0393s    Reduced Chi^2 = 4.0403      Levenberg-Marquardt = 0.1
 Free parameter values:
         33.23         0.7286          35.21        0.03152
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.6506s    Reduced Chi^2 = 3.7095      Levenberg-Marquardt = 0.01
 Free parameter values:
          48.7           0.98          35.58       0.007004
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.2928s    Reduced Chi^2 = 3.7335      Levenberg-Marquardt = 0.001
 Free parameter values:
         45.93         0.9825          35.75       0.008385
 Total time = 5.723s    Reduced Chi^2 = 3.7084      Levenberg-Marquardt = 0.01
 Free parameter values:
            46         0.9837          35.76       0.008304
-------------------------------------------------------------------------------
Fit converged:

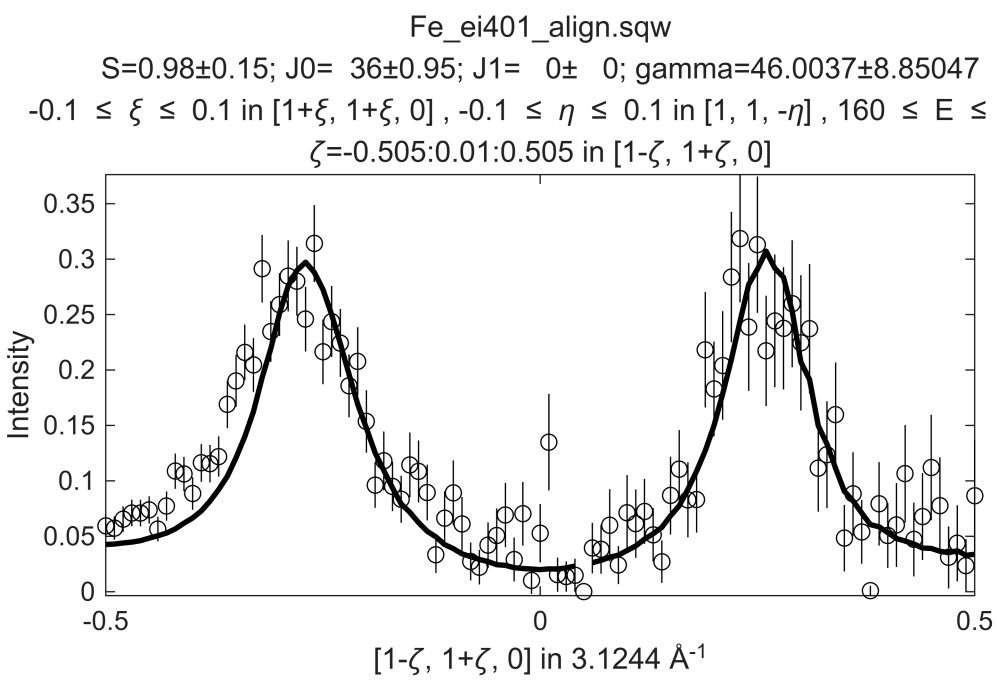
Parameter values (free parameters with error estimates):
-------------------------------------------------------

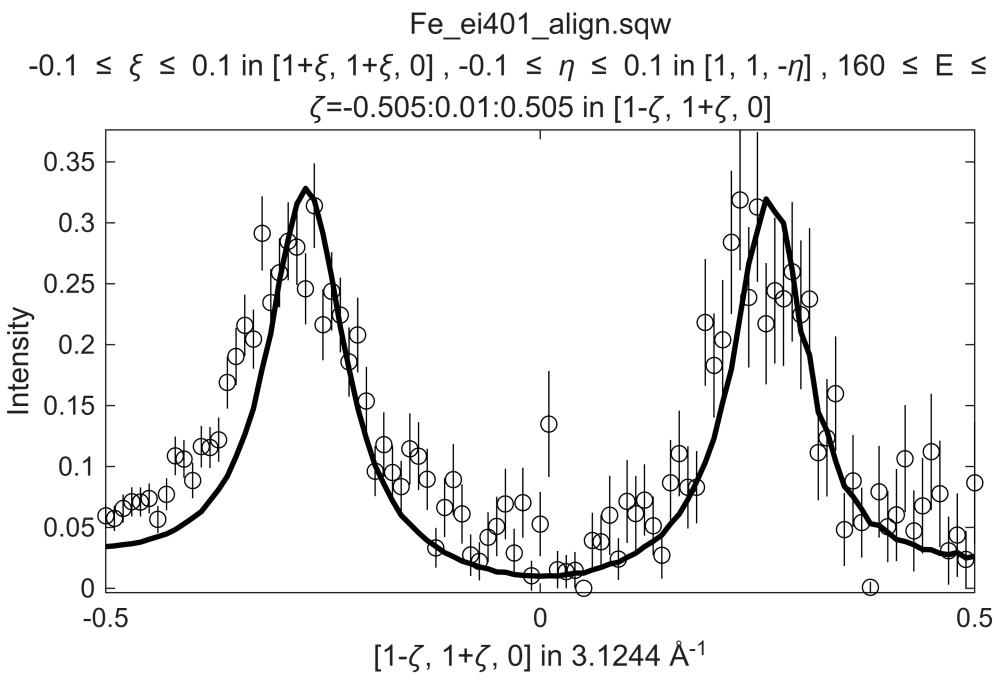
Foreground parameter values:
----------------------------
    1              1
    2              8
    3             46   +/-   8.85
    4         0.9837   +/-   0.1486
```

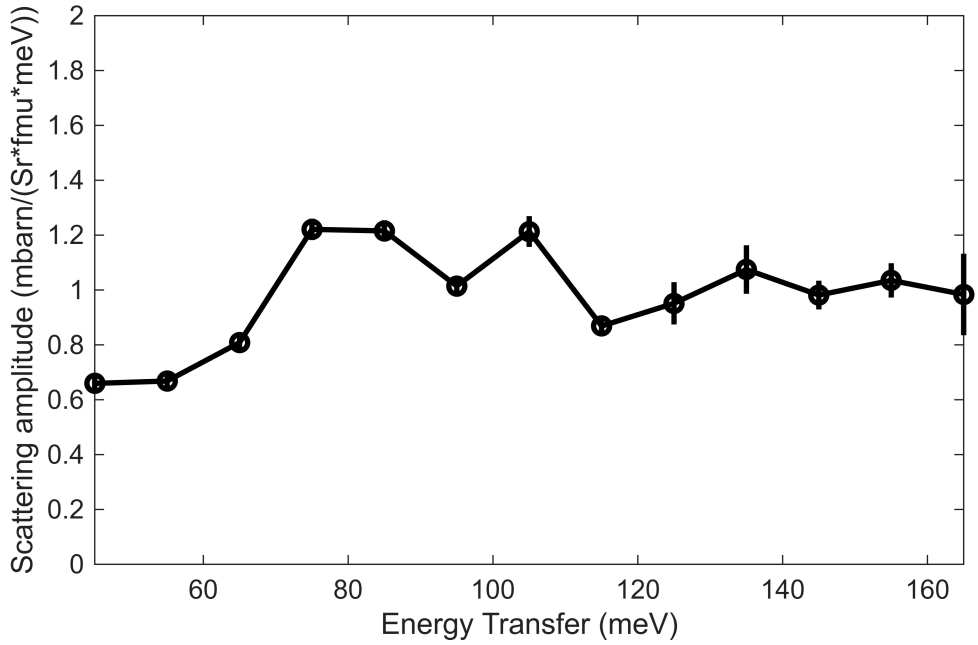
```
     5              0
     6          35.76   +/-   0.9514
     7              0
     8              0
     9              0
    10              0

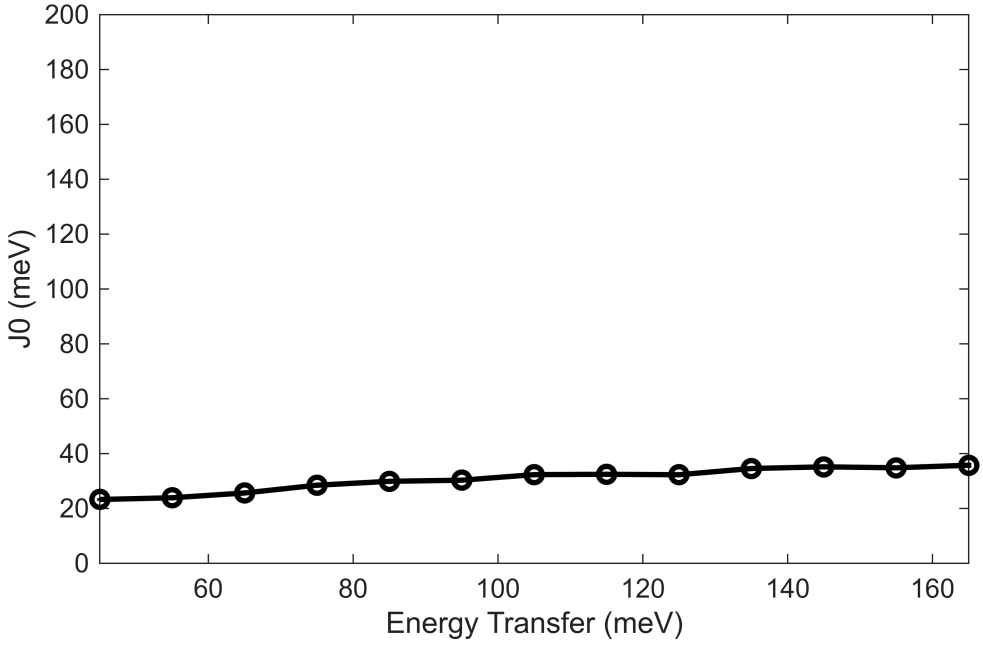

Background parameter values:
----------------------------
     1       0.008304   +/-   0.01022
     2              0
     3              0

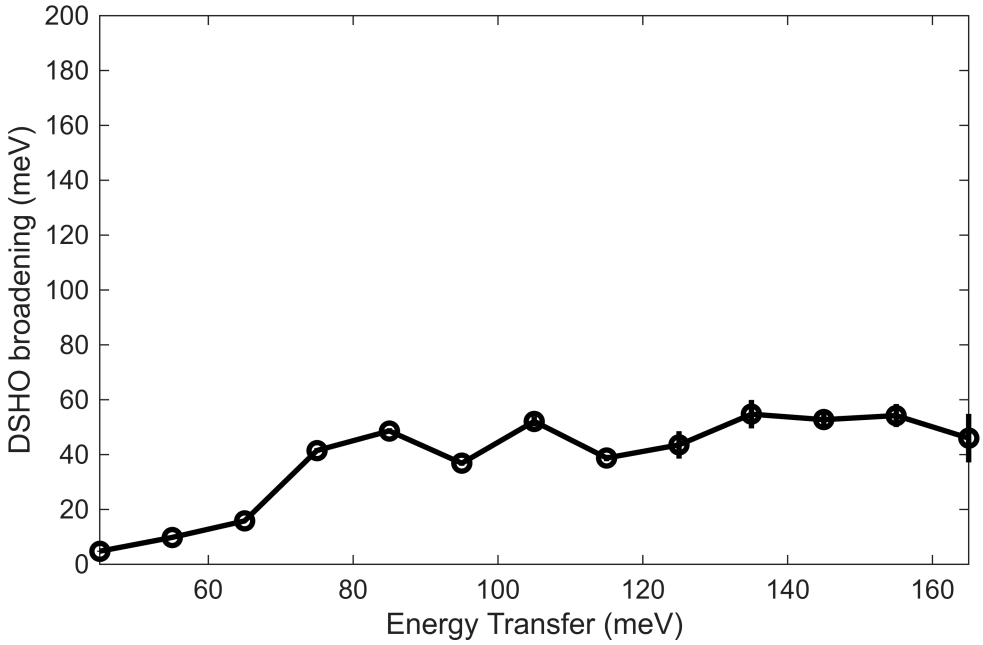

Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.8834    0.7318   -0.8120
    0.8834    1.0000    0.6785   -0.8961
    0.7318    0.6785    1.0000   -0.4612
   -0.8120   -0.8961   -0.4612    1.0000
```






Scattering amplitude (mbarn/(Sr*fmu*meV))
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
60
80
100
120
140
160
Energy Transfer (meV)

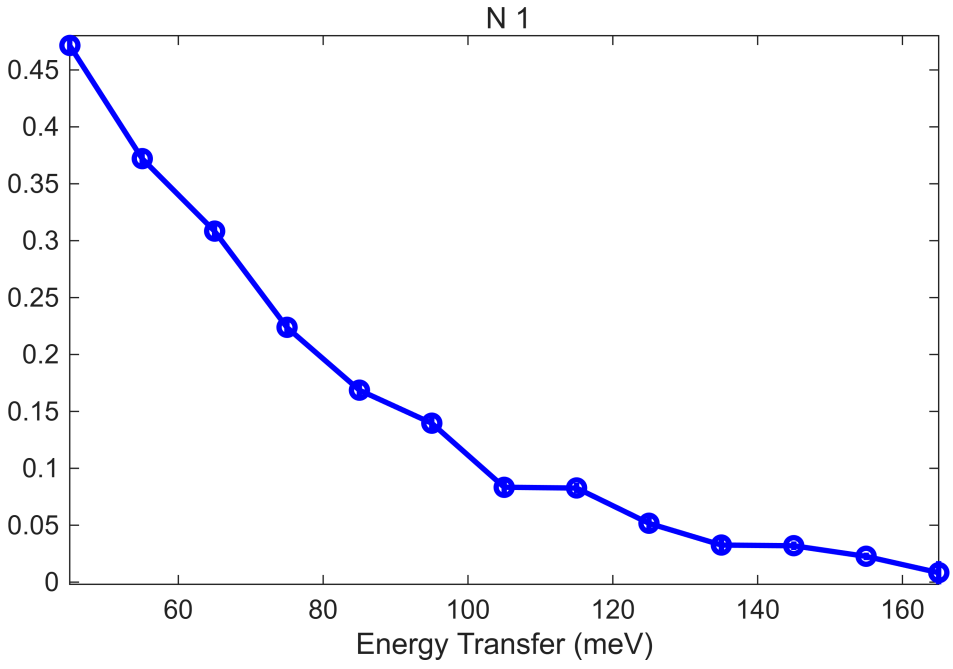

J0 (meV)
200
180
160
140
120
100
80
60
40
20
0
60
80
100
120
140
160
Energy Transfer (meV)



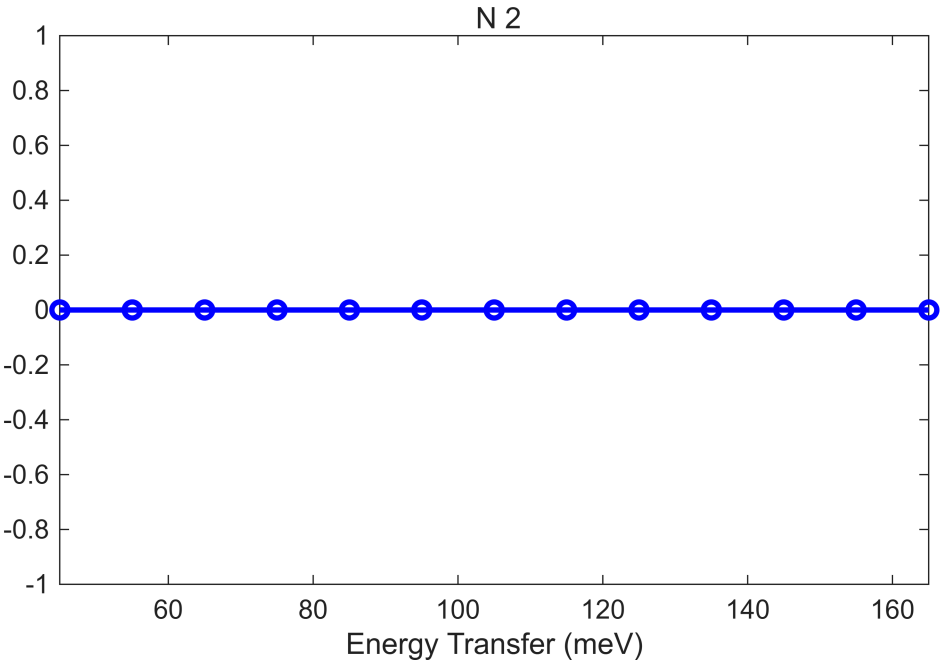

```
cl = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep
```

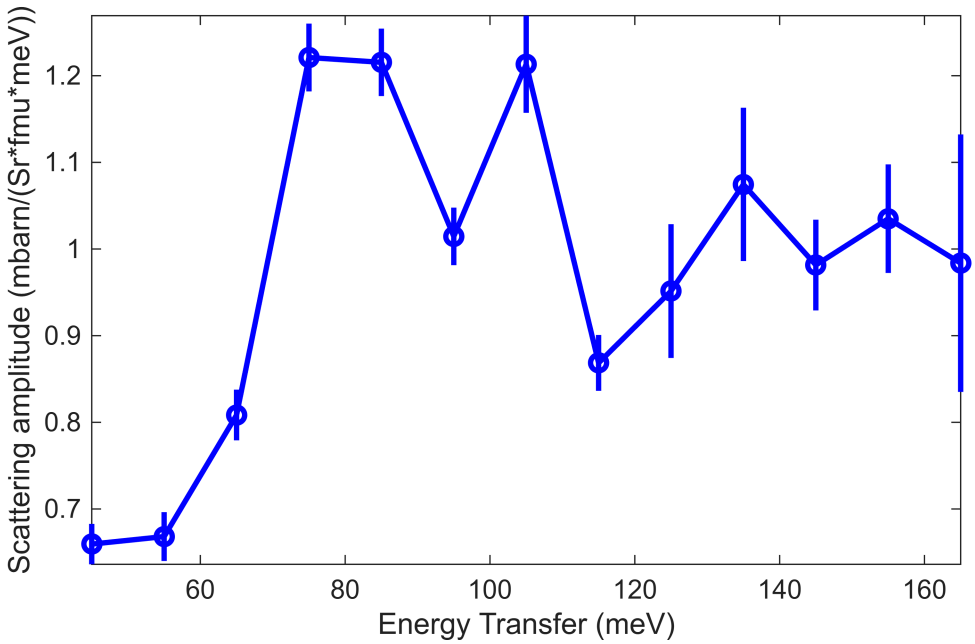
```
all_fit_par = NG_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
acolor b
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en); keep_figure;
```



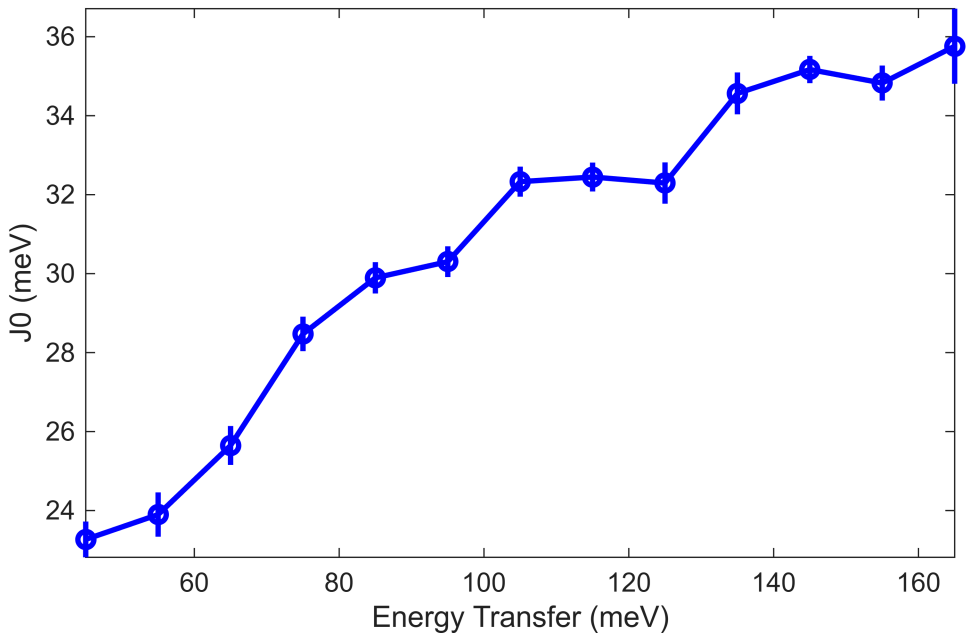
```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



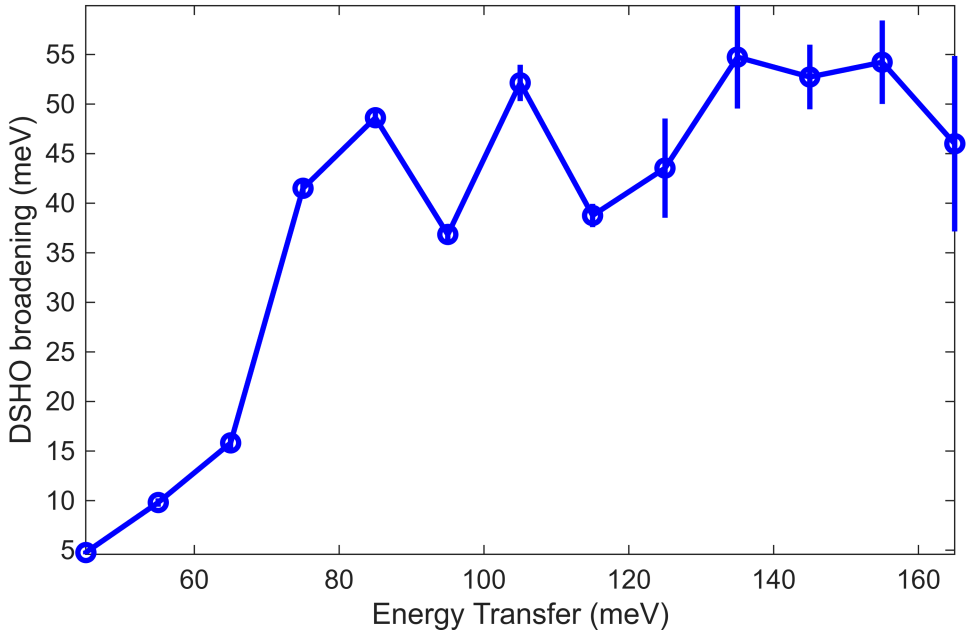

```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
[S_eff,J0_eff,G_eff] = extract_fit_par(all_fit_par );
pd(S_eff); keep_figure;
```




```
pd(J0_eff);keep_figure
```




```
pd(G_eff); keep_figure;
```




```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struc = NG_FFCutsFits;
save(CutFitFileNG,"fit_src_struc",'-v7.3')
```